

Note: The Practice Book is for reference purpose, LIU Test question paper may not be compulsory set from this practice book.

Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
1	1	Which of the following functions can be used to fill all null values in a data frame?	A	1		fillna()	filled()	fillnull()	filler()
2	1	Which attribute of dropna() can be used to select the columns from which null values are to be considered for removing rows?	C	1	LIU 2025	thresh	how	subset	superset
3	1	Which of the following pandas functions is used to generate cross tabulation?	D	1		crosstabulation	cross_tabulation	cross_tab	crosstab
4	1	Which of the following DataFrame attributes is used to return one or more specified row(s)?	C	1		locate	location	loc	find
5	1	Which of the following attributes can be used to show the number of rows and columns in a Pandas dataframe?	D	1		size	info	describe	shape
6	1	Which of the following is not displayed by the Pandas DataFrame info function?	D	1		column names	non-null count	data types	column average
7	1	Which of the following is not displayed by the Pandas DataFrame describe function?	C	1		count	mean	correlation	std
8	1	What does it indicate if the corr() function shows correlation as 1 between any two columns of the DataFrame?	A	1		perfect correlation	good correlation	bad correlation	none of these
9	1	Which of the following is an example of qualitative data?	B	1		mean	gender	median	mode
10	1	Which of the following is an example of quantitative data?	B	1	LIU 2025	eye colour	weight	skin colour	names
11	1	What does DataFrame.dropna(how='all') do?	A	1		drops those rows from the DataFrame which contain all null values	drops all rows from the DataFrame	drops even numbered rows from the DataFrame	drops odd numbered rows from the DataFrame
12	1	A data point that differs significantly from other observations is known as	D	1		mean	median	mode	outlier
13	1	Which of following pandas functions can be used to display the specified number of rows from the beginning of the dataset?	A	1		head()	tail()	begin()	end()
14	1	Which of following pandas functions can be used to display the specified number of rows from the end of the dataset?	B	1		head()	tail()	begin()	end()
15	1	Which of the following represents each data sample as polyline connecting parallel lines where each parallel line represents an attribute of that data sample?	A	1		parallel coordinates	parallelogram	straight lines	long lines
16	1	What is the output of the code shown below? import pandas as pd import numpy as np df=pd.DataFrame([[0,1,0,2,0,np.nan,5],[2,0,0,1,0,5,0,np.nan],[5,0,0,1,0,np.nan,5,0]]) df.dropna() print(df.loc[1,3])	B	1	LIU 2025	0	5,0	1.0	2.0
17	1	What is the output of the code shown below? import pandas as pd import numpy as np df=pd.DataFrame([[0,1,2,np.nan,5],[2,0,1,5,np.nan],[5,0,1,np.nan,5],[2,0,1,np.nan,np.nan]]) df=df.drop_duplicates(subset=[1,2]) df=df.drop_duplicates(subset=[4]) df.dropna(thresh=2,axis=1) print(df.shape)	C	1		(5, 2)	(2, 3)	(2, 5)	(3, 2)
18	1	What type of Error the following code produces? import pandas as pnd pnd.Series([1,2], index= ['a','b','c'])	D	1		Syntax	Index	Key	Value
19	1	To remove multiple values from the Pandas dataframe and to keep only the first occurrence values, what will be the correct syntax?	C	1		df.drop_duplicate()	df.drop()	df.drop_duplicates()	df.dropduplicates()
20	1	From a Pandas series 's', if we need to extract indices (1,5,7,12) what will be the syntax used?	B	1		s (1,5,7,12)	s ([1,5,7,12])	s.index([1,5,7,12])	s.index(1,5,7,12)
21	1	What is the method used to calculate the mean of a numeric column in a DataFrame ?	D	1		average()	calculate_mean()	get_mean()	mean()

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22	1	What is the output of the below code? import pandas as pd import numpy as np df=pd.DataFrame({"a":[1,2,np.nan,3,4],"b":[1,5,np.nan,2,1]}) df=df.drop_duplicates(subset="b") df.dropna() df.fillna(20,inplace=True) print(df.shape[0])	A	1	LJU 2023	4	3	2	1
23	1	What is the output of the below code? import pandas as pd import numpy as np df=pd.DataFrame([[1,2,3,4,5],[2,1,3,4,5],[np.nan,np.nan,np.nan,np.nan,np.nan]]) df.dropna(thresh=3,axis=1,inplace=True) print(df.shape[1])	A	1	LJU 2023	0	1	2	3
24	1	What is the output of the below code? import pandas as pd import numpy as np df=pd.DataFrame([[1,2,3,4,5],[2,1,3,4,5],[np.nan,np.nan,np.nan,np.nan,np.nan]]) df.drop(1,inplace=True) df=df.dropna() print(df.shape[0])	A	1	LJU 2023	1	2	3	0
25	1	What will be the output of following code? import pandas as pd data = { "Student": ["A","B","C","D","E"], "Math": [78, 90, 67, 88, 90], "Science": [85, 92, 70, 88, 85] } df = pd.DataFrame(data) result = df.sort_values(by=["Math","Science"], ascending=[False,True]) print(result.head(1)["Student"].values[0])	D	1		A	B	D	E
26	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "B": [1,2], "A": [3,4], "C": [5,6] }) df2 = df.sort_index(axis=1) print(df2.columns[0])	A	1		A	B	C	NONE OF ABOVE
27	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Price": [100,200,150], "Qty": [2,3,4] }) df[["Total"]] = df.apply(lambda x: x["Price"]*x["Qty"], axis=1) print(df[["Total"]].sum())	C	1		1000	1100	1400	1300
28	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Marks": [45.7, 78.2, 90.9] }) df[["Marks"]] = df[["Marks"]].astype(int) print(df[["Marks"]].sum())	A	1		213	214	215	216
29	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "City": ["A","B","C","A","B"] }) df2 = df.replace({"A": "X", "B": "Y"}) print(df2[["City"]].unique())	B	1		['A','B','C']	['X','Y','C']	['X','Y']	ERROR

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30	1	What will be the output of following code? import pandas as pd df1 = pd.DataFrame({"A":[1,2]}) df2 = pd.DataFrame({"B":[3,4]}) result = pd.concat([df1, df2], axis=0) print(result.shape)	B	1		(2,2)	(4,2)	(4,1)	(2,1)
31	1	What will be the output of following code? import pandas as pd df1 = pd.DataFrame({"A":[1,2]}) df2 = pd.DataFrame({"A":[3,4]}) result = pd.concat([df1,df2], ignore_index=True) print(result.index[3])	B	1		2	3	4	ERROR
32	1	What will be the output of following code? import pandas as pd df1 = pd.DataFrame({ "ID":[1,2,3], "Name":["A","B","C"] }) df2 = pd.DataFrame({ "ID":[2,3,4], "Marks":[80,90,70] }) result = pd.merge(df1, df2, how="inner", on="ID") print(len(result))		1		1	2	3	4
33	1	What will be the output of following code? import pandas as pd df1 = pd.DataFrame({ "ID":[1,2,3], "Name":["A","B","C"] }) df2 = pd.DataFrame({ "ID":[2], "Marks":[80] }) result = pd.merge(df1, df2, how="left", on="ID") print(result["Marks"].isnull().sum())	B	1		1	2	3	0
34	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Team":["X","Y","X","Y","X"], "Runs":[50,60,70,80,90] }) result = df.groupby("Team")["Runs"].mean() print(result["X"])	D	1		70	75	80	NONE OF ABOVE
35	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Dept":["IT","HR","IT","HR","IT","Finance"] }) print(df.groupby("Dept").size()["IT"])	B	1		2	3	4	5
36	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Category":["A","A","B","B"], "Value":[10,20,30,40] }) result = df.groupby("Category").agg({"Value":["sum","max"]}) print(result.loc["B",("Value","sum")])	D	1		30	40	60	70
37	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "City":["A","A","B","C","C","C"] }) print(df.groupby("City").nunique().shape[0])	B	1		2	3	4	6

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38	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Team":["X","X","Y","Y"], "Score":[10,20,30,40] }) result = df.groupby("Team").nth(1) print(result.loc["X"])	D	1		10	20	40	KeyError
39	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Age":[22,25,30,35,40], "Salary":[30000,40000,50000,60000,70000] }) result = df[(df["Age"]>25) & (df["Salary"]>=50000)] print(len(result))	B	1		2	3	4	1
40	1	What will be the output of following code? import pandas as pd import numpy as np df = pd.DataFrame({ "A":[1,2,3], "B":[4,5,6] }) result = df.apply(sum, axis=0) print(result["A"])	B	1		3	6	9	12
41	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Team":["A","A","B","B"], "Score":[10,20,30,40] }) result = df.groupby("Team").sample(1) print(len(result))	B	1		1	2	3	4
42	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Dept":["IT","HR","IT","Finance"] }) groups = df.groupby("Dept").groups print(groups["IT"])	D	1		0	1	2	NONE OF ABOVE
43	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Name":["A","B","C","D","E"], "Score":[90,90,85,90,85], "Age":[20,22,21,19,23] }) result = df.sort_values(by=["Score","Age"], ascending=[False,True]).reset_index(drop=True) print(result.loc[1, "Name"])	A	1		A	B	D	C
44	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "C":[1,2], "A":[3,4], "B":[5,6] }) df = df.drop("B", axis=1) df2 = df.sort_index(axis=1) print(list(df2.columns))	B	1		['C','A']	['A','C']	['B','C']	ERROR
45	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({"A":[1.9,2.1,3.7]}) df["A"] = df["A"].astype(int) print(df["A"].mean())	A	1		2.0	2.3	2.6	3

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46	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({"A": [1,2,3,1,2]}) df2 = df.replace({1:100,2:200}) print(df2["A"].sum())	C	1		403	503	603	703
47	1	What will be the output of following code? df1 = pd.DataFrame({"A": [1,2]}, index=[0,1]) df2 = pd.DataFrame({"A": [3,4]}, index=[1,2]) result = pd.concat([df1,df2]) print(result.loc[2]["A"])	c	1		2	3	4	ERROR
48	1	What will be the output of following code? import pandas as pd df1 = pd.DataFrame({"A": [1,2,3]}) df2 = pd.DataFrame({"B": [4,5]}) result = pd.concat([df1,df2], axis=1) print(result.isnull().sum().sum())	B	1		0	1	2	3
49	1	What will be the output of following code? import pandas as pd df1 = pd.DataFrame({"ID": [1,2], "Score": [80,90]}) df2 = pd.DataFrame({"ID": [1,2], "Score": [85,95]}) result = pd.merge(df1,df2,on="ID",suffixes=("_X", "_Y")) print(result.loc[0,"Score_Y"])	B	1		80	85	90	95
50	1	What will be the output of following code? import pandas as pd df1 = pd.DataFrame({"ID": [1,2,3]}) df2 = pd.DataFrame({"ID": [2,3,4]}) result = pd.merge(df1,df2,on="ID",how="outer",indicator=True) print((result["_merge"]=="both").sum())	C	1		0	1	2	ERROR
51	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Dept": ["IT", "IT", "HR"], "Salary": [100,200,300] }) result = df.groupby("Dept",as_index=False).sum() print(type(result))	C	1		Series	dict	DataFrame	list
52	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Cat": ["A","A","B","B"], "Val": [5,15,10,20] }) result = df.groupby("Cat").agg({"Val": ["min", "max", "mean"]}) print(result.loc["A",("Val", "mean")])	B	1		5.0	10.0	15.0	20.0
53	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Team": ["X", "X", "Y"], "Score": [10,20,30] }) result = df.groupby("Team").apply(lambda x: x["Score"].max()-x["Score"].min()) print(result["X"])	C	1		0	5	10	15

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54	1	What will be the output of following code? import pandas as pd df = pd.DataFrame({ "Dept":["IT","IT","HR"], "City":["A","A","B"] }) result = df.groupby("Dept").nunique() print(result.loc["IT","City"])	A	1		1	2	3	4																																																											
55	1	Create a Pandas DataFrame from the following table and write code to remove all rows from this table containing at least one NaN value <table><tr><th></th><th>name</th><th>region</th><th>sales</th><th>expenses</th></tr><tr><td>0</td><td>William</td><td>NaN</td><td>50000.0</td><td>42000.0</td></tr><tr><td>1</td><td>Emma</td><td>North</td><td>52000.0</td><td>43000.0</td></tr><tr><td>2</td><td>Sofia</td><td>East</td><td>NaN</td><td>NaN</td></tr><tr><td>3</td><td>Markus</td><td>NaN</td><td>NaN</td><td>NaN</td></tr><tr><td>4</td><td>Edward</td><td>West</td><td>42000.0</td><td>38000.0</td></tr><tr><td>5</td><td>Thomas</td><td>West</td><td>72000.0</td><td>39000.0</td></tr><tr><td>6</td><td>Ethan</td><td>South</td><td>49000.0</td><td>42000.0</td></tr><tr><td>7</td><td>NaN</td><td>NaN</td><td>NaN</td><td>NaN</td></tr><tr><td>8</td><td>Arun</td><td>West</td><td>67000.0</td><td>39000.0</td></tr><tr><td>9</td><td>Anika</td><td>East</td><td>65000.0</td><td>50000.0</td></tr><tr><td>10</td><td>Paulo</td><td>South</td><td>67000.0</td><td>45000.0</td></tr></table>		name	region	sales	expenses	0	William	NaN	50000.0	42000.0	1	Emma	North	52000.0	43000.0	2	Sofia	East	NaN	NaN	3	Markus	NaN	NaN	NaN	4	Edward	West	42000.0	38000.0	5	Thomas	West	72000.0	39000.0	6	Ethan	South	49000.0	42000.0	7	NaN	NaN	NaN	NaN	8	Arun	West	67000.0	39000.0	9	Anika	East	65000.0	50000.0	10	Paulo	South	67000.0	45000.0		3				
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58	1	Write Python code to remove outliers from any given DataFrame.		4																																																																

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59	1	Consider the following data: data = { "A": ["TeamA", "TeamB", "TeamB", "TeamC", "TeamA"], "B": [50, 40, 40, 30, 50], "C": [True, False, False, False, True] } Convert this to a Pandas DataFrame and remove duplicate rows from it. Reset index values.		4					
60	1	Consider the following autmpg dataset: https://raw.githubusercontent.com/Jovita7/Data-Analysis-and-Visualization/main/auto-mpg.csv Write Python code to convert it to a DataFrame and remove mpg and cylinders columns from it		3					
61	1	Use the file heights_weights.csv (https://raw.githubusercontent.com/Jovita7/Data-Analysis-and-Visualization/main/heights_weights.csv) which contains 10000 non-null values for heights and weights. The Male column shows 1 if the person is a Male and 0 if the person is a Female. 1. Convert this file into a pandas Data Frame. (0.5 marks) 2. Display basic information like memory and data types for this data frame. (0.5 marks) 3. Display basic statistics like mean, std, quartiles, etc. for this data frame. (0.5 marks) 4. Create a correlation table for the data frame and comment about what kind of correlation is there between Height and Weight. (0.5 marks) 5. Do Height and Weight contain any outliers? (1 mark)		3					
62	1	Use the file ipl-matches.csv which contains data of all the IPL matches from year 2008 to 2022. Read this csv file and display the basic information like memory and data types for this data frame. Write python code for the following cases: 1. List out all matches gone in superover. 2. How Many Matches won by Chennai Super Kings at Kolkata. 3. In How Many Matches MS Dhoni is Player of Match Vs Mumbai Indians. 4. Display list of all matches in which Gujarat Titans won the Toss and Elected to Bat and won the match. 5. Display list of all matches won by Gujarat Titans.		5					
63	1	Use the file spotify.csv 1.Convert this file into a pandas Data Frame. (0.5 marks) 2.Display basic information like memory and data types for this data frame. (0.5 marks) 3.Display basic statistics like mean, std, quartiles, etc. for this data frame. (0.5 marks) 4.Create a correlation table for the data frame and comment about what kind of correlation is there between danceability and energy (0.5 marks) 5.Display first five rows for this data frame.(0.5 marks) 6.Display last five rows for this data frame. (0.5 marks) 7.Display the rows between 15 to 39 for this data frame. (0.5 marks) 8.Display the data only for last five rows and last five columns for this data frame. (0.5 marks) 9.Display the shape for this data frame. (0.5 marks) 10.Display the sum of NULL values for all the columns. (0.5 marks) 11.Remove first 3 columns from this Data Frame. (0.5 marks) 12.Remove first 10 rows from this Data Frame. (0.5 marks) 13.After removing first 3 columns and first 10 rows from this data frame find outliers for the column popularity. (1 marks) 14.After removing first 3 columns and first 10 rows from this data frame remove outliers for the column energy then display the data frame. (1 marks) 15.Display cross tabulation between time_signature and track_genre for actual Data Frame. (1 marks)		9					
64	1	Read this csv file car data.csv and display the basic information like memory and data types for this data frame. Write Python Code for the following cases: 1.How Many ritz car are there with kms driven more than 30000km. 2.How many Petrol cars are of the year 2017 and whose selling price > 10 lakhs. 3.How many swift cars are there with selling price < 4 Lakhs. 4.How many Automatic Transmission Petrol Car are of the year 2015 whose selling price is > 10 Lakhs. 5.List out Vehicles with Automatic Transmission and selling price < 1 Lakh. 6.How Many Petrol Vehicles are there with kms driven more than 50000kms and Year is between 2010 to 2015(both Year included). 7.List out the cars whose price difference between present price and selling price is > 15 lakhs. 8.How many Petrol vehicles are there whose kms driven < 5000km and selling price < 50000.		9	LIJ 2024				

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
65	1	1. Load the dataset into a pandas DataFrame (data_result.csv) and answer the following questions. 2. View the first few rows of the dataset 3. Check the shape of the dataset 4. View the first last rows of the dataset 5. Get summary statistics of numerical columns 6. Get summary statistics of numerical columns with 0.58 and 0.87 percentiles 7. Get summary statistics of all types of columns 8. Information of all columns 9. Check for missing values 10. Removing duplicates if duplicates 11. List out female students who have greater than 7 spi in all semesters. 12. Find number of students those who have greater than 8 spi in all 5 semesters. 13. Find outliers of sem 4 result. Also represent statistical analysis with visualization.(boxplot)		9					
66	1	Use the file movies.csv which contains 1629 rows and 18 columns. Read this csv file and display the basic information like memory and data types for this data frame. Write python code for the following cases: 1.List out Movies Released in Year 2019. 2.How Many Movies are having IMDB Rating > 7 (Display Number of Movies). 3.List out the Movies with 'title' and 'story' whose IMDB Votes > 20000. 4.List out Movies Released in Year 2018, Display only Movie Title with Release Date of Year 2018 Movies. 5.Display only Movie Title with its Wikipedia Link.		6					
67	1	You are given a CSV file named ipl-matches.csv, which contains data of all Indian Premier League (IPL) matches played from the year 2008 to 2022. Read the dataset and display the basic information about the DataFrame. Using Python and Pandas, write code to answer the following questions: 1.How many matches were won by Gujarat Titans in a Super Over ? 2.How Many Matches won by Royal Challengers Bangalore Vs Kolkata Knight Riders. 3.How many Matches won by Toss Winner of the Match. 4.In how many matches was the result decided by the D/L method and the match was won by wickets? 5.In how many matches Toss won by Rajasthan Royals & Won the Match. 6.In how many matches V Kohli declared player of the match against Chennai Super Kings. 7.Which Team is the Toss Winner of the Final match Played in Dubai? 8.List out the Finals played by Kolkata Knight Riders in which Toss Won by Kolkata Knight Riders in the Finals and choose to field.		5	LUJ 2025				
68	1	import pandas as pd data = { "EmpID": [101,102,103,104,105], "Dept": ["IT", "HR", "IT", "Finance", "HR"], "Salary": [60000,45000,80000,75000,50000], "Age": [25,32,29,41,28] } df = pd.DataFrame(data) Sort employees by Dept (Ascending) and Salary (Descending). After sorting, sort the index in descending order. Display the final DataFrame.		5					
69	1	import panda as pd df = pd.DataFrame({ "Student": ["A", "B", "C", "D"], "Math": [70,45,90,60], "Science": [75,40,85,55] }) Create a column Average using apply() row-wise. Create a column Grade based on: ≥ 80 → Distinction ≥ 60 → First Otherwise → Second Display final DataFrame.		5					

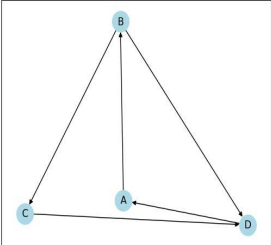
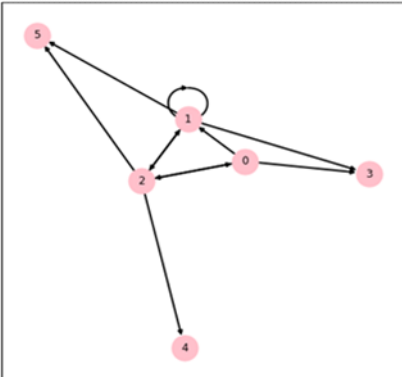
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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
70	1	import pandas as pd df = pd.DataFrame({ "Marks":["50","60","Absent","70","Absent"] }) Replace "Absent" with 0. Convert the column to integer type. Print mean marks. Print number of students scoring above 50.		5					
71	1	import pandas as pd df1 = pd.DataFrame({ "ID":[1,2,3], "Name":["A","B","C"] }) df2 = pd.DataFrame({ "ID":[2,3,4], "Marks":[80,90,70] }) Concatenate row-wise using outer join. Count total missing values. Sort the result by ID.		5					
72	1	import pandas as pd students = pd.DataFrame({ "ID":[1,2,3,4], "Name":["A","B","C","D"] }) marks = pd.DataFrame({ "ID":[2,3,5], "Score":[85,90,75] }) Perform outer merge using indicator=True. Count:		5					
73	1	import pandas as pd df = pd.DataFrame({ "Dept":["IT","IT","HR","HR","Finance","Finance"], "Salary":[50000,70000,40000,60000,80000,75000] }) Group by Dept. Calculate mean, median, and max salary using agg().		5					
74	1	import pandas as pd df = pd.DataFrame({ "Team":["A","A","A","B","B"], "Runs":[50,70,80,40,60] }) Calculate range (max - min) per team using groupby + apply.		5					
75	1	import pandas as pd df = pd.DataFrame({ "Salary":[20000,22000,21000,25000,24000,300000] }) Detect outliers using IQR method. Remove outliers. Print cleaned dataset		5					
76	1	import pandas as pd df = pd.DataFrame({ "Dept":["IT","IT","HR","HR","Finance","Finance"], "Employee":["A","B","C","D","E","F"], "City":["X","X","Y","Z","X","Y"] }) Find number of unique cities per department. Take one random employee per department. Display results.		5					

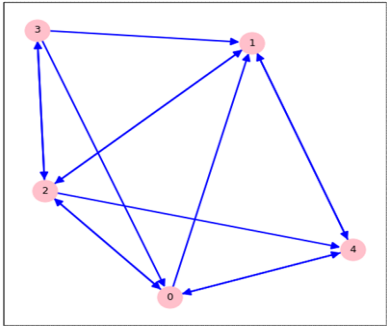
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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
77	1	import pandas as pd import pandas as pd df = pd.DataFrame({ "Product":["P1","P2","P3","P4","P5"], "Price":[100,200,150,300,250], "Quantity":[5,2,4,1,3] }) Create a new column TotalAmount = Price × Quantity using apply() row-wise. Sort the DataFrame by TotalAmount in descending order.		5					
78	1	import pandas as pd df = pd.DataFrame({ "Employee":["A","B","C","D","E"], "Dept":["IT","HR","IT","Finance","HR"], "Salary":[50000,40000,70000,80000,45000] }) Group employees by Department. Calculate total salary and average salary per department. Display departments where average salary is greater than 50,000.		5					
79	1	import pandas as pd df = pd.DataFrame({ "Dept":["IT","IT","HR","HR","Finance","Finance"], "Employee":["A","B","C","D","E","F"], "Experience":[2,5,3,7,4,6] }) Find the number of employees per department using size(). Find the employee with highest experience in each department using nth() or sorting. Display results.		5					
80	2	Which of the following is best suitable chart to show data correlation?	D	1		Histogram	Bar	Pie	Scatter
81	2	What type of data is best suited for box plots?	C	1		Categorical data	Binary data	Continuous numerical data	Time-series data
82	2	In a box plot, the bottom line of the box represents which quartile?	A	1	LIJ 2023	First quartile	Second quartile	Third quartile	Fourth quartile
83	2	In a box plot, the top line of the box represents which quartile?	C	1		First quartile	Second quartile	Third quartile	Fourth quartile
84	2	What is a heatmap used for?	B	1		To visualize categorical data	To visualize numerical data in a grid-like format	To fit a regression line to the data	To perform clustering on the data
85	2	Which parameter in the sns.heatmap() function is used to show numerical values in heatmap?	A	1	LIJ 2025	annot	annotate	percent	show
86	2	What is the purpose of the cbar parameter in the sns.heatmap() function?	C	1		To adjust the transparency of the colorbar	To adjust the size of the colorbar	To add a colorbar to the heatmap	To adjust the color scale of the heatmap
87	2	Which of the following types of graphs is not supported by NetworkX?	C	1		Directed graphs	Undirected graphs	Hypergraphs	None of the above
88	2	Which of the following methods is used to add nodes to a graph in NetworkX?	A	1	LIJ 2025	graph.add_node()	graph.add_nodes()	graph.nodes()	graph.node()
89	2	Which parameter in sns.heatmap() displays numeric values inside each cell?	B	1		cmap	annot	cbar	linewidths
90	2	If linewidths=2 and linecolor='black' are set in heatmap, what happens?	B	1		Data values increase	Cell borders become thick black lines	Heatmap becomes darker	Colorbar disappears
91	2	Which parameter removes the color bar in heatmap?	B	1		annot=False	cbar=False	vmax=0	linewidths=0
92	2	In sns.boxplot(), what does hue parameter do?	B	1		Changes box width	Groups boxplots by category	Changes y-axis	Removes outliers
93	2	Which parameter controls color in sns.boxplot()?	B	1		cmap	color	annot	linewidths
94	2	In sns.scatterplot(), style parameter controls what?	B	1		Marker color	Marker shape	Marker size	Axis label
95	2	Which parameter sets marker size in sns.scatterplot()?	C	1		size	markers	s	linewidth
96	2	What is the purpose of vmin and vmax in heatmap?	B	1		Filter data	Fix color scaling range	Remove negative values	Change axis labels
97	2	Which barmode creates grouped bars in px.bar()?	C	1		orientation	text_auto	group	color
98	2	Which barmode is used for stacked bars in px.bar()?	C	1		group	relative	stack	overlay
99	2	Which barmode is used for diverging bars px.bar()?	C	1		stack	group	relative	none
100	2	What does pattern_shape parameter do in px.bar()?	B	1		Changes bar height	Adds pattern based on category	Adds tooltip	Changes axis
101	2	orientation='h' in px.bar() creates what?	B	1		Vertical bars	Horizontal bars	3D bars	Stacked bars
102	2	Which function creates a 3D scatter plot?	C	1		px.scatter()	px.scatter_matrix()	px.scatter_3d()	px.line()
103	2	Which parameter controls bubble size in px.scatter()?	A	1		size	width	symbol	markers
104	2	symbol parameter in scatter plot controls what in px.scatter()?	C	1		Marker color	Marker size	Marker shape	Axis range

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Sr. No.	unit_number	Question_text	answer	marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
105	2	Which function creates scatter matrix?	C	1		px.scatter()	px.scatter_3d()	px.scatter_matrix()	px.histogram()
106	2	facet_col in scatter plot is used for what in px.scatter()?	B	1		Change marker	Create multiple subplots	Add legend	Change axis
107	2	Which parameter shows markers in line chart?	B	1		hover_data	markers=True	text_auto	color
108	2	How to plot multiple lines in px.line()?	A	1		Use color parameter	Use list in SIZE parameter	Use symbol parameter	Use orientation
109	2	color_discrete_sequence is used for what IN px.line()?	B	1		Change axis	Set custom line colors	Remove legend	Set marker size
110	2	Which parameter controls number of bins in px.histogram()?	A	1		nbins	histfunc	color	width
111	2	histfunc='sum' in histogram does what in px.histogram()?	B	1		Count frequency	Calculate sum per bin	Remove bins	Change color
112	2	Which method shows percent and label in pie chart?	B	1		update_layout()	update_traces(textinfo='percent+label')	text_auto=True	color_discrete_map
113	2	Which heatmap parameter controls color palette?	B	1		annot	cmmap	cbar	vmin
114	2	Setting annot=False in heatmap will:	A	1		Hide values	Hide color	Hide grid	Hide axis
115	2	In sns.boxplot(), x parameter usually represents:	B	1		Numeric value	Category variable	Color map	Marker size
116	2	Which seaborn function creates scatter plot?	C	1		sns.boxplot()	sns.heatmap()	sns.scatterplot()	px.scatter()
117	2	In px.bar(), color parameter is used to:	B	1		Change bar height	Group bars by category	Change axis	Change orientation
118	2	px.bar() width parameter controls:	B	1		Bar width	Figure width	Axis width	Legend width
119	2	In px.scatter(), symbol parameter is used to:	B	1		Change color	Change marker shape	Change size	Change axis
120	2	Which function creates stacked histogram?	A	1		px.histogram()	px.bar()	px.line()	px.pie()
121	2	In px.histogram(), color parameter does what?	A	1		Groups data by category	Changes bin size	Removes bins	Adds line
122	2	Which function creates simple line chart?	A	1		px.line()	px.bar()	px.scatter()	px.histogram()
123	2	markers=True in px.line() will:	A	1		Add markers on data points	Remove line	Change color	Hide legend
124	2	Which parameter sets custom colors in px.pie()?	A	1		color_discrete_sequence	text_auto	hover_data	pattern_shape
125	2	In px.pie(), values parameter represents:	B	1		Category labels	Numeric values	Color sequence	Legend title
126	2	scatter_matrix is used to:	B	1		Create 3D plot	Compare multiple numeric variables	Create histogram	Create pie chart
127	2	Which plotly function supports 3D plotting?	B	1		px.scatter()	px.scatter_3d()	px.line()	px.bar()
128	2	Which parameter adds extra information on hover?	A	1		hover_data	text_auto	pattern_shape	orientation
129	2	text_auto=True does what?	A	1		Adds automatic value labels	Changes color	Changes size	Removes legend
130	2	Which parameter in px.scatter() creates subplots by row?	A	1		facet_row	facet_col	symbol	size
131	2	Which seaborn function supports hue parameter?	A	1		sns.scatterplot()	sns.heatmap()	Both	None
132	2	In heatmap, linewidths controls:	B	1		Color intensity	Cell border thickness	Axis title	Legend
133	2	Which plot type shows proportion of total?	C	1		Bar	Line	Pie	Scatter
134	2	px.scatter_matrix() requires which parameter?	A	1		dimensions	color_discrete_map	nbins	barmode
135	2	In px.bar(), text parameter is used for:	A	1		Manual label display	Color control	Size control	Axis control
136	2	Write a python program which creates following graph using networkx module in python 		2					
137	2	Using Networkx Library, Write a Code to create below given Directed Graph. 		4	LIJ 2025				

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
138	2	Create a boxplot of the distribution of temperatures in different cities. Take data from 'temperatures.csv' from below: https://raw.githubusercontent.com/kavit88/Data-Sets/main/temperatures.csv		3					
139	2	The following dictionary shows how five people follow each other on Instagram: instagram = {'person1': [0,1,1,0,1], 'person2': [0,0,1,0,1], 'person3': [1,1,0,1,1], 'person4': [1,1,1,0,0], 'person5': [1,1,0,0,0]} E.g., the list for person1 has the value on index 2 as 1 which means person1 followsperson3 and a directed edge should be added from person1 to person3. Using networkx library, create a directed graph.		4					
140	2	You have been given a dataset of car prices and their respective horsepower, mileage, and weight. You have been tasked to analyze the relationship between these variables and create a scatter plot to visualize the patterns. Dataset: The dataset, named "car_data.csv" : https://raw.githubusercontent.com/kavit88/Data-Sets/main/car_data.csv		5					
141	2	You have been given a dataset of house prices and their respective lot size and square footage. Your task is to create a scatter plot to determine if there is any correlation between these variables. Dataset: The dataset, named "house_data.csv": https://raw.githubusercontent.com/kavit88/Data-Sets/main/house_data.csv		5					
142	2	Use the file heights_weights.csv which contains 10000 non-null values for heights and weights. The Male column shows 1 if the person is a Male and 0 if the person is a Female. Take file of dataset from: https://raw.githubusercontent.com/kavit88/Data-Sets/main/heights_weights.csv 1. Convert this file into a pandas Data Frame. 2. Display basic information like memory and data types for this data frame. 3. Display basic statistics like mean, std, quartiles, etc. for this data frame. 4. Create a correlation table for the data frame and comment about what kind ofcorrelation is there between Height and Weight. 5. Do Height and Weight contain any outliers? Answer by creating boxplots for both. 6. Finally, create a scatter plot of Weight v/s Height with the following specifications:		6					
143	2	The file "student_scores.csv" contains the marks scored by a group of students in three subjects: Maths, Science, and English. Each row contains the name of the student, their score in Maths, Science, and English. Create a pandas DataFrame from this data and create a heatmap to visualize the correlations between the scores in these three subjects. Take Dataset from below: https://raw.githubusercontent.com/kavit88/Data-Sets/main/student_scores.csv		3					
144	2	Using Networkx Library, Write a code to Create a below given Directed Graph. 		4	LUJ 2024				

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
145	2	You have been hired as a network analyst by a company to analyze the social network of their employees. The company has provided you with the following data: There are 5 employees in the company, each identified by a unique ID from 1 to 5. The following relationships exist between the employees: 1. Employee 1 is friends with Employee 2 and Employee 3. 2. Employee 2 is friends with Employee 4. 3. Employee 3 is friends with Employee 5. Your task is to create a NetworkX graph representing this social network and display it.		3	LJU 2023				
146	2	You have been hired by an Airlines company to analyze their routes. The company has provided you following data. Your task is to create a NetworkX directed graph representing the routes and display it. Figure size should be (15,15), node color should be green, take appropriate node size, edge color should be red. Data: Kolkata to Mumbai Mumbai to Pune Mumbai to Goa Kolkata to Delhi Kolkata to Bhubaneswar Mumbai to Delhi Delhi to Chandigarh Delhi to Surat Kolkata to Hyderabad Hyderabad to Chennai Chennai to Thiruvananthapuram Thiruvananthapuram to Hyderabad Kolkata to Varanasi Delhi to Varanasi Mumbai to Bangalore Chennai to Bangalore Hyderabad to Bangalore Kolkata to Guwahati		4					
147	2	Using 'supermarket_sales.csv' file do the following operations and give required answer by using proper programming process. 1). Load the dataset into a pandas DataFrame and read first 8 rows. 2). Check for missing values and fill it by mean values of that particular column if any. 3). Find the number of orders which have 'Quantity' less than 3 and which have (either 'Rating' greater than 8.5 or 'Total' greater than 600). 4). Find the sum of 'Total' purchasing price spent by Member and Normal 'Customer type'. 5). Find the percentage of total of 'gross income' based on the different 'Payment' methods used by customers. (Ewallet, Cash and Credit card) 6). Analyze the purchasing behavior of male and female customers using 'Gender' column. Find their average purchase prices using 'Total' column. 7). Create a scatter plot that shows the relationship between total amount spent and rating. 8). Create a box plot that shows the distribution of 'Rating' and 'Quantity'. And comment about outliers in both columns. 9). Visualize with parallel co-ordinates for 'Unit price', 'Total', 'cogs' columns' data with respect to 'Product line'.		9					
148	2	Use the file data.csv which contains 169 rows and 4 columns. 1. Convert this file into pandas Data Frame and Display basic statistics like mean, std, quartiles, etc. for this data frame. 2. Create a correlation table for the data frame and comment about what kind of correlation is there between Duration and Calories? 3. Find whether there any null or NA values, drop all such rows if found in the data frame and print the shape of the data frame after dropping. 4. Prepare a scatter matrix for the following data frame and prepare a parallel coordinates for Duration v/s Pulse, Maxpulse and Calories (all 3 other columns). 5. Do Maxpulse have any outliers? Find using function. 6. Show the outliers using box plot for Maxpulse, width of box plot should be 0.75 and notch should be True. 7. Create a scatter plot for Duration (x-axis) and then Pulse, Maxpulse and Calories (y-axis) with different colors. For each there should be different color and marker.		9					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
149	2	For the given dataset – iris.csv, perform following exploratory data analysis using python - Use comment feature to answer appropriate questions – a) Load dataset into jupyter notebook using appropriate libraries. Check the datatypes of the dataset attributes. Does the data contain any missing /null values? b) Extract head and tail of the dataset using appropriate methods. c) Summarize statistical figures (i.e. mean, median, percentiles) in one table using appropriate method. d) Create correlation table of all variables. What can you infer about relation between petal length and sepal length? e) Create parallel coordinate plot of iris dataset. What can you infer about petal length and petal width? f) Create box plot of sepal width. Visualizing the plot, answer whether the sepal width data contains any outliers. g) Create cross tabulation of sepal length and petal width attributes. What does the table represent? h) Create scatter matrix of the dataset. i) Create a new column called 'Sepal.Length.Size' which contains "High" if sepal length ≥ 5 or "Low" if sepal length < 5 .		9					
150	2	To upload the 'diabetes_unclean.csv' to your working folder First import the following libraries import pandas as pd import matplotlib.pyplot as plt import numpy as np 1.Make a data frame with the variable name df 2.To display the specific statistics or measures that are relevant for object-type columns 3.To display the specific statistics or measures that are relevant for numerical-type columns 4.How many rows and columns are in a given dataset 5.To check the missing values 6.To replace the missing values in the column "HbA1c" with their mean value 7.Dropping the missing values of other columns 8.Display the correlation between variables 9. Checking the outliers in the dataset for the following parameters: 'AGE', 'Urea', 'HbA1c', 'Chol', 'TG', 'HDL', 'LDL', 'VLDL', 'BMI' using box plot with labels and title 10.Visualized the "Urea", "HbA1c", "TG" and "BMI" parameters for different ages using parallel_coordinates with labels and title 11.Remove the rows whose gender column has an "F" value and give the frequency count of the "F" and "M" values in different CLASS values 12.Remove the outliers in the "HbA1c" columns and print the shape of the data frame Note: all task output with specific question numbers and follow the sequence Example: print("Ans-1")		9	UJU 2023				
151	2	import pandas as pd import seaborn as sns import matplotlib.pyplot as plt df = pd.DataFrame({ "Math":[78,85,90,66,88], "Science":[80,82,89,70,85], "English":[75,88,92,60,84], "Computer":[90,91,95,72,89] }, index=["S1","S2","S3","S4","S5"]) Plot heatmap,,annot=True,cmap="coolwarm", linewidths=1, linecolor="black" Show colorbar		5					
152	2	import pandas as pd df = pd.DataFrame({ "Height":[150,160,170,180,175], "Weight":[50,60,65,75,70], "Age":[20,22,25,30,28], "BMI":[22,24,23,26,25] }) Compute correlation, Plot heatmap with cmap="YlGnBu"		5					
153	2	import pandas as pd df = pd.DataFrame({ "Dept":["IT","IT","HR","HR","Finance","Finance"], "Salary":[50000,70000,40000,60000,80000,75000], "Gender":["M","F","M","F","M","F"] }) Plot boxplot of Salary vs Dept with hue=Gender.		5					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
154	2	import pandas as pd df = pd.DataFrame({ "Salary":[20000,22000,21000,23000,25000,300000] }) Plot boxplot and observe outlier.		5					
155	2	import pandas as pd df = pd.DataFrame({ "StudyHours":[2,3,5,6,8], "Marks":[50,60,75,85,95], "Gender":["M","F","M","F","M"] }) Create dataset of students with Marks and StudyHours. Use sns.scatterplot() Use hue for Grade. Use style for Gender.		5					
156	2	Simple Bar Chart: Visualize total revenue of different departments. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Department": ["IT", "HR", "Finance", "Marketing"], "Revenue": [120000, 80000, 150000, 95000] })		5					
157	2	Grouped Bar Chart: Compare yearly sales across regions. Compare yearly sales across regions. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Region": ["North", "North", "South", "South"], "Year": ["2023", "2024", "2023", "2024"], "Sales": [200, 250, 180, 220] })		5					
158	2	Diverging Bar Chart: Display profit and loss of departments. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Department": ["IT", "HR", "Finance", "Marketing"], "Profit_Loss": [50000, -20000, 70000, -15000] })		5					
159	2	2D Group Scatter: Analyze relationship between Height and Weight grouped by Gender. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Height": [150, 160, 170, 180, 165], "Weight": [50, 60, 70, 80, 65], "Gender": ["M", "F", "M", "F", "M"] })		5					
160	2	Bubble Scatter: Sales vs Profit where bubble size represents Quantity. df = pd.DataFrame({ "Sales": [100, 200, 300, 400], "Profit": [20, 50, 70, 90], "Quantity": [10, 30, 50, 80] })		5					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
161	2	3D Scatter:Visualize 3D coordinates with color grouping. df = pd.DataFrame({ "X":[1,2,3,4], "Y":[10,20,30,40], "Z":[5,15,25,35], "Category":["A","B","A","B"] })		5					
162	2	single Line Chart:Show yearly company growth. df = pd.DataFrame({ "Year":[2020,2021,2022,2023,2024], "Revenue":[100,150,200,250,300] })		5					
163	2	line Chart with Color Group: Track product sales trends. df = pd.DataFrame({ "Month":["Jan","Feb","Mar","Jan","Feb","Mar"], "Sales":[50,60,70,40,55,65], "Product":["A","A","A","B","B","B"] })		5					
164	2	Line Chart with Color Group: Track product sales trends. df = pd.DataFrame({ "Month":["Jan","Feb","Mar","Jan","Feb","Mar"], "Sales":[50,60,70,40,55,65], "Product":["A","A","A","B","B","B"] })		5					
165	2	Student Marks Distribution: Create a dataset of 20 students containing their exam marks and grade category (A, B, C). Perform the following: Plot a histogram of Marks. Set nbins = 5. Use color = "Grade". Display values using text_auto = True. Add an appropriate title. import pandas as pd import plotly.express as px import plotly.io as pio # Set notebook renderer pio.renderers.default = "notebook" # Create Dataset df = pd.DataFrame({ "Student": [f"S{i}" for i in range(1, 21)], "Marks": [45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 48, 52, 58, 62, 68, 72, 78, 82, 88, 92], "Grade": ["C","C","C","B","B","B","A","A","A","A", "C","C","C","B","B","B","A","A","A","A"] })		5					
166	2	Create dataset of monthly sales transactions (30 values) and Region (North, South). Perform: Plot histogram of Sales. Use color="Region". Use pattern_shape="Region". Set nbins=6. Add title and custom color sequence. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Sales":[100,120,150,180,200,220,250,270,300,320, 110,130,160,190,210,230,260,280,310,330, 105,125,155,185,205,225,255,275,305,325], "Region":["North"]*15 + ["South"]*15 })		5					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
167	2	Employee Age Distribution:Create dataset of employee ages and departments. Perform: Plot histogram of Age. Use color="Department". Use histfunc="count". Set nbins=5. Add proper title and width=700, height=400. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Age":[22,25,28,30,35,40,45,50,29,33, 26,31,38,42,48]. "Department":["IT","IT","HR","HR","Finance","Finance", "IT","HR","Finance","IT", "HR","Finance","IT","HR","Finance"] })		5					
168	2	Product Market Share:Create dataset showing sales percentage of 5 products. Perform: Plot pie chart using px.pie(). Show percentage and label. Use custom color sequence. Add title. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Product":["A","B","C","D","E"], "Sales":[30,25,20,15,10] })		5					
169	2	Department Budget Allocation: Create dataset of department budget allocation. Perform: Plot pie chart. Pull the highest budget department outward. Show percentage and label. Set text position inside. import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "Department":["IT","HR","Finance","Marketing"], "Budget":[50000,30000,40000,20000] })		5					
170	2	Population Share by City:Create dataset of population of 4 cities. Perform: Plot pie chart. Use custom color sequence. Pull second slice outward. Display percent and label.import pandas as pd import plotly.express as px import plotly.io as pio pio.renderers.default = "notebook" df = pd.DataFrame({ "City":["City A","City B","City C","City D"], "Population":[400,300,200,100] })		5					
171	3	Which of the following pandas functions is used to convert categorical data into numeric data?	A	1		get_dummies()	numeric()	get_categorical()	get_dumps()

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
172	3	How do you handle missing or corrupted data in a dataset?	D	1		Drop missing rows or columns	Replace missing values with mean/median/mode	Assign a unique category to missing values	All of these
173	3	What is Scikit-learn?	A	1		A machine learning library in Python	A data visualization library in Python	A natural language processing library in Python	A web development framework in Python
174	3	Which of the following is an example of a regression algorithm in Scikit-learn?	C	1		K-means clustering	Decision tree	Linear regression	Support vector machines (SVM)
175	3	How would you access the column "symboling" from the dataframe df?	A	1		df["symboling"]	df=="symboling"	df[:"symboling"]	df[["symboling"]]
176	3	What is the correct symbol for missing data?	B	1	LJU 2024	na	nan	none	non
177	3	Why do we convert values of Categorical Variables into numerical values?	A	1		Most statistical models cannot take in objects or strings as inputs	To save memory	To save time	None of these
178	3	What is the main difference between regression and classification in supervised learning?	A	1		Regression predicts continuous outcomes, while classification predicts categorical outcomes	Regression predicts categorical outcomes, while classification predicts continuous outcomes	Regression uses labeled data, while classification uses unlabeled data	Regression is unsupervised, while classification is supervised
179	3	What evaluation metric is commonly used for regression tasks?	C	1		Accuracy	Precision	Mean Squared Error (MSE)	Recall
180	3	What type of target variable is typically used in a regression problem?	C	1	LJU 2024, LJU 2025	Discrete	Categorical	Continuous	Binary
181	3	What is feature selection in supervised learning?	B	1	LJU 2024	It is the process of creating new features from existing ones.	It is the process of removing irrelevant or redundant features from the dataset.	It is the process of selecting the target variable for prediction.	It is the process of transforming categorical features into numerical features.
182	3	What is feature transformation in machine learning?	D	1		It is the process of creating new features from existing ones.	It is the process of removing irrelevant or redundant features from the dataset.	It is the process of selecting the target variable for prediction.	It is the process of transforming categorical features into numerical features.
183	3	What is the main purpose of train-test split?	B			Increase dataset size	Evaluate model on unseen data	Reduce features	Speed up training
184	3	Which sklearn function is used for train-test split?	C			split_data()	divide()	train_test_split()	data_partition()
185	3	Typical training data percentage is:	C			20%	50%	70-80%	100%
186	3	The remaining data after training is used for:	B			Cleaning	Testing	Scaling	Encoding
187	3	Which parameter defines test data size?	B			split_size	test_size	ratio	size
188	3	What does random_state ensure?	C			More accuracy	Faster execution	Reproducibility	Less data
189	3	If test_size=0.3, training size is:	C			0.3	0.5	0.7	1.0
190	3	You've been given a dataset with apartment area and price information. There's a noticeable non-linear relationship between area and price. To address this you intend to categorize them into 'High', 'Medium', and 'Low' groups. Prices above \$3,000,000 are 'High', below \$2,000,000 are 'Low', and between \$2,000,000 and \$3,000,000 are 'Medium'. Write a code to achieve this assuming that dataset has two columns named area and price.		3	LJU 2025				
191	3	In a survey dataset, you have a column representing participants' ages. You want to categorize ages into 'Young', 'Middle-aged', and 'Elderly' groups. Ages below 30 are 'Young', ages between 30 and 60 are 'Middle-aged', and ages above 60 are 'Elderly'. Write a code to achieve this assuming the dataset has a column named 'age'.		3					
192	3	In a customer dataset, you have a column representing customer incomes. You want to categorize incomes into 'Low', 'Medium', and 'High' groups. Incomes below 30000 are 'Low', incomes between 30000 and 70000 are 'Medium', and incomes above 70000 are 'High'. Write a code to achieve this assuming the dataset has a column named 'income'.		3					
193	4	From where you can import LinearRegression?	C	1		sklearn.metrics	sklearn.linearmodel	sklearn.linear_model	sklearn.model_selection
194	4	From where you can import train_test_split?	D	1		sklearn.metrics	sklearn.linearmodel	sklearn.linear_model	sklearn.model_selection
195	4	What is the purpose of the predict() method in sklearn?	B	1		To train a model using a given dataset	To make predictions using a trained model	To evaluate the performance of a model	To split the data in train and test data
196	4	What is the purpose of the fit() method in sklearn?	A	1		To train a model using a given dataset	To evaluate the performance of a model	To create a plot of predicted values	All of these
197	4	If we pass x and y to a function train_test_split(), we will get output in which order?	B	1		x_train, y_train, x_test, y_test	x_train, x_test, y_train, y_test	x_train, y_test, x_test, y_train	y_train, y_test, x_train, x_test
198	4	Consider the following lines of code, what is the name of the column that contains the target values: from sklearn.linear_model import LinearRegression lm=LinearRegression() X = df[['highway-mpg']] Y = df['price'] lm.fit(X, Y) Yhat=lm.predict(X)	A	1	LJU 2025	price	highway-mpg	Both A and B	None of these

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
199	4	If X is a dataframe with 100 rows and 5 columns, and y is the target with 100 samples, and assuming all the relevant libraries and data have been imported, and the following line of code has been executed: LR = LinearRegression() LR.fit(X, y) yhat = LR.predict(X) How many samples does yhat contain?	C	1		50	500	100	5
200	4	What will be the size of training data if data is split like below? train_test_split(x,y,test_size=0.25,random_state=2)	A	1		0.75	0.25	0.8	0.2
201	4	Consider the following code snippet that implements linear regression in Python, what will be printed as the output of the code snippet? import numpy as np from sklearn.linear_model import LinearRegression # Training data X_train = np.array([[1], [2], [3], [4]]) y_train = np.array([2, 4, 6, 8]) # Test data X_test = np.array([[5]]) # Linear regression model model = LinearRegression() model.fit(X_train, y_train) predicted_value = model.predict(X_test) print(predicted_value)	B	1	UJU 2024	10	[10,]	5	[5,]
202	4	If the data contains 100 rows and 2 columns and if test_size=0.2 then how many rows will go into training and how many will undergo in testing?	A	1		80,20	70,30	50,60	30,70
203	4	Consider the following lines of code having 200 non-null data in both x and y. what is the output of following code : import pandas as pd import numpy as np dataset=pd.read_csv("advertising.csv") x=dataset[["TV","Radio","Newspaper"]] y=dataset["Sales"] print(x.shape)	A	1	UJU 2025	(200,3)	(200,)	(200,1)	(160,3)
204	4	Consider the following lines of code having 300 non-null data in both x and y. what is the output of following code : import pandas as pd import numpy as np dataset=pd.read_csv("Book1.csv") x=dataset[["cgpa"]] y=dataset["package"] from sklearn.model_selection import train_test_split x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.2, random_state=1) print(x_train.shape)	C	1	UJU 2024	(240,3)	(240,)	(240,1)	(60,)
205	4	If a dataframe with 400 rows and 5 columns, from the following code how many number of rows will go for x_test? from sklearn.model_selection import train_test_split x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.2, random_state=1)	A	1		80	100	10	200
206	4	In scikit-learn's linear regression, what is the purpose of the "coef_" attribute?	B	1		It returns the intercept of the linear regression model.	It provides the coefficients of the features in the linear regression model.	It predicts the target variable values for new input data.	It computes the mean squared error (MSE) of the model.
207	4	What is the purpose of the LinearRegression() function in scikit-learn?	B	1		To perform classification tasks	To fit a linear model to the data	To preprocess text data	To plot scatter plots
208	4	In linear regression, what does the coefficient of determination (R-squared) measure?	D	1		The strength of the relationship between independent and dependent variables	The slope of the regression line	The accuracy of the model predictions	The variance explained by the regression model
209	4	When should you use linear regression for modeling data?	D	1		When the relationship between variables is nonlinear	When the dataset contains categorical variables	When the dependent variable is binary	When there is a linear relationship between independent and dependent variables
210	4	What does the coefficient of the independent variable (slope) in a simple linear regression model represent?	B	1		The y-intercept of the regression line.	The change in the dependent variable for a unit change in the independent variable.	The standard deviation of the residuals.	The correlation between independent and dependent variables
211	4	In polynomial regression, what does the degree of the polynomial represent?	B	1		The number of independent variables in the model.	The order of the polynomial curve fitted to the data.	The correlation between independent and dependent variables	The y-intercept of the regression curve.
212	4	What distinguishes polynomial regression from linear regression?	C	1		Polynomial regression can handle categorical variables.	Polynomial regression only works with two variables.	Polynomial regression fits a curve to the data instead of a straight line.	Polynomial regression always has a higher R-squared value than linear regression.

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																																
213	4	When would you choose polynomial regression over linear regression?	B	1		When there is a linear relationship between variables.	When the data points exhibit a non-linear pattern.	When dealing with categorical variables.	When the dataset contains missing values.																																
214	4	Match the following concepts/methods in regression with their correct descriptions: <table><tr><th>Concept/Method</th><th>Description</th></tr><tr><td>A. Simple Linear Regression</td><td>1. Transformation used to create polynomial features from the original features.</td></tr><tr><td>B. R-Score</td><td>2. Model that fits a linear relationship between the target variable and a single predictor variable.</td></tr><tr><td>C. PolynomialFeatures</td><td>3. Metric used to evaluate the proportion of variance in the dependent variable explained by the model.</td></tr><tr><td>D. train_test_split</td><td>4. Method to divide the dataset into training and testing subsets.</td></tr><tr><td>E. LinearRegression</td><td>5. Model that can fit both simple and multiple linear regression.</td></tr><tr><td>F. Polynomial Regression</td><td>6. Scenario where a model learns the noise in the training data and performs poorly on new data.</td></tr><tr><td>G. Overfitting</td><td>7. Model that fits a non-linear relationship between the target variable and predictor variables by using polynomial terms.</td></tr></table>	Concept/Method	Description	A. Simple Linear Regression	1. Transformation used to create polynomial features from the original features.	B. R-Score	2. Model that fits a linear relationship between the target variable and a single predictor variable.	C. PolynomialFeatures	3. Metric used to evaluate the proportion of variance in the dependent variable explained by the model.	D. train_test_split	4. Method to divide the dataset into training and testing subsets.	E. LinearRegression	5. Model that can fit both simple and multiple linear regression.	F. Polynomial Regression	6. Scenario where a model learns the noise in the training data and performs poorly on new data.	G. Overfitting	7. Model that fits a non-linear relationship between the target variable and predictor variables by using polynomial terms.	A	1	LJU 2024	A-2,B-3,C-1,D-4,E-5,F-7,G-6	A-5,B-2,C-3,D-1,E-7,F-4,G-6	A-5,B-3,C-2,D-4,E-1,F-7,G-6	A-2,B-3,C-4,D-1,E-7,F-6,G-5																
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215	4	For x = np.array([5, 15, 25, 35, 45, 55]) and y = np.array([5, 20, 14, 32, 22, 38]), apply simple linear regression using scikit learn library and calculate calculate R squared, coefficient and intercept. Predict the y values for x = np.arange(5). (Don't split data for training/testing)		3																																					
216	4	Given a dataset with 'SAT' scores as independent variables and 'GPA' as the dependent variable, calculate R squared, coefficient and intercept using linear regression and scikitlearn library. (Don't split data for training/testing)		3																																					
217	4	Given a real estate price size year dataset, implement multiple linear regression using scikitlearn library. Using the model, make a prediction about an apartment price with size 750 sq.ft. for 2009.Also Calculate R squared, coefficient and intercept. (Don't split data for training/testing)		3																																					
218	4	Predict salary based on job position of 6.5 using polynomial regression with a degree of 3 and scikit learn library for the given 'Position_Salaries.csv' dataset. (Don't split data for training/testing)		3																																					
219	4	For x = np.arange(0, 30) and y = np.array([3, 4, 5, 7, 10, 8, 9, 10, 10, 23, 27, 44, 50, 63, 67, 60, 62, 70, 75, 88, 81, 87, 95, 100, 108, 135, 151, 160, 169, 179]), apply polynomial regression using scikit learn library and calculate R squared, coefficient and intercept. Predict the y values for x = np.arange(5). (Don't split data for training/testing)		3																																					
220	4	Write a program to make a model based on linear regression for the following dataframe created from a csv file named "Package.csv" of x and y which follows equation y = a + bx. Write a program which can predict value of y based on any value of x, also write code to find value of a and b in above equation. Given Data in csv file: <table><tr><th>cgpa</th><th>package</th></tr><tr><td>6.89</td><td>3.26</td></tr><tr><td>5.12</td><td>1.98</td></tr><tr><td>7.82</td><td>3.25</td></tr><tr><td>7.42</td><td>3.67</td></tr><tr><td>6.94</td><td>3.57</td></tr><tr><td>7.89</td><td>2.99</td></tr><tr><td>6.73</td><td>2.60</td></tr><tr><td>6.75</td><td>2.48</td></tr><tr><td>6.09</td><td>2.31</td></tr><tr><td>8.31</td><td>3.51</td></tr><tr><td>5.32</td><td>1.86</td></tr><tr><td>6.61</td><td>2.60</td></tr><tr><td>8.94</td><td>3.65</td></tr><tr><td>6.93</td><td>2.89</td></tr><tr><td>7.73</td><td>3.42</td></tr></table>	cgpa	package	6.89	3.26	5.12	1.98	7.82	3.25	7.42	3.67	6.94	3.57	7.89	2.99	6.73	2.60	6.75	2.48	6.09	2.31	8.31	3.51	5.32	1.86	6.61	2.60	8.94	3.65	6.93	2.89	7.73	3.42		3					
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221	4	Write a program to make a model based on linear regression for the following dataframe created from a csv file named "data.csv" of x1 and y which follows equation y = a+bx1. Write a program which can predict value of y based on any value of x, also write code to find value of a and b in above equation. Given Data in csv file:		3	LJU 2024																																				

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																		
		<table><tr><td>y</td><td>X₁</td></tr><tr><td>140</td><td>60</td></tr><tr><td>155</td><td>62</td></tr><tr><td>159</td><td>67</td></tr><tr><td>179</td><td>70</td></tr><tr><td>192</td><td>71</td></tr><tr><td>200</td><td>72</td></tr><tr><td>212</td><td>75</td></tr><tr><td>215</td><td>78</td></tr></table>	y	X ₁	140	60	155	62	159	67	179	70	192	71	200	72	212	75	215	78							
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179	70																										
192	71																										
200	72																										
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222	4	Write a program to create a Model using linear regression to predict the price of house using the csv file provided named "Housing.csv". Do the required process in the data before making a model. Find predicted values, co-efficients, intercept and mean squared error. https://github.com/pdsinroza/python2/blob/39b36bf2f0121910fd1207952aa0ec20b2d77cfb/housing.csv		6																							
223	4	Write a program to create a Model using linear regression to predict the student scores using the csv file provided named "student_scores.csv". Do the required process in the data before making a model. Find predicted values, co-efficients, intercept and mean squared error. https://github.com/pdsinroza/python2/blob/695586ff85947e2ff727385ce208322f5b29de08/student_scores.csv		6																							
224	4	Write a program to create a Model using linear regression to predict the gas consumption using the csv file provided named "petrol_consumption.csv". Do the required process in the data before making a model. Find predicted values, co-efficients, intercept and mean squared error. https://github.com/pdsinroza/python2/blob/f4711a48cc10c84c9892b96900760848e1c1fdf0/petrol_consumption.csv		6																							
225	4	Write a program to create a Model using linear regression to predict the gas consumption using the csv file provided named "FuelConsumptionCo2.csv". Do the required process in the data before making a model. Find predicted values, co-efficients, intercept and mean squared error. (Wherever required remove null values, convert categorical data into numeric data) (Print Output wherever required)		6																							
226	4	For the given RealEstate csv, write a python program satisfying following tasks to demonstrate application of machine learning through multiple linear regression as follows – Given: - Dataset RealEstate.csv ML Library to be used scikit-learn Dependent variable 'Y house price of unit area' Independent variables 'X1 transaction date', 'X2 house age', 'X3 distance to the nearest MRT station', 'X4 number of convenience stores', 'X5 latitude' and 'X6 longitude' 1. Import required libraries. 2. Load RealEstate dataset, create a dataframe and check datatypes of its attributes using appropriate method. 3. Remove 'No' column from the dataframe. 4. Check for any null values in features using appropriate method. 5. Create feature variables x and y as given above. 6. Create training and testing sets of feature variables with 70% of data for training and with random state of 110. 7. Create and fit regression model using appropriate method. 8. Use testing set created in step 6 to find and print the prediction of the outcome. 9. Find and print coefficient and mean squared error of the regression model.		7																							
227	4	Write a program to create a Model using linear regression to predict the charges of insurance using the csv file provided named "insurance.csv". Do the required process in the data before making a model. Find predicted values, co-efficients, intercept and mean squared error.		5																							
228	4	Write a program to create a Model using linear regression to predict the wine quality using the csv file provided named "winequality.csv". Do the required process in the data before making a model. If you find any null value in "winequality.csv" then replace null value with mean value of respected columns. Find co-efficient, intercept and mean squared error. also Predict the quality of red wine for the following data: fixed acidity: 8 volatile acidity: 0.4 citric acid: 0.40 residual sugar: 15 chlorides: 0.048 free sulfur dioxide: 40 total sulfur dioxide: 150 density: 0.99 pH: 3 sulphates: 0.45 alcohol: 10.5		5																							
229	4	Consider variables x and y created from a pandas dataframe "car.csv" . Create new column named "Age_car" (Age_car=2023-year) For multiple linear regression problem, x contains the independent variables (Age_car , Driven_kms , Fuel_Type , Selling_type , Transmission) and y contains the dependent (Selling_Price) variable which is to be predicted. Write a Python program to split x and y into training and testing datasets with a 20% split. Then create a multiple linear regression model using the training data and print its coefficients ,intercept and mean squared error.		9	LUJ 2024																						

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Sr. No.	unit_number	Question_text	answer	marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
230	4	Write a python program for following tasks - Make sure all the required libraries are imported at the beginning of the Jupyter Notebook. a)Load the windpower dataset into the notebook. b)Check records for any missing values and remove them permanently using appropriate method. c)Treat windspeed as the independent variable and power as the dependent variable. Split the dataset into training (75%) and testing (25%) sets using random state of 42. d)Use scikit-learn to train polynomial regression models of degrees 3, 4, 5, and 6. e)Predict power values on the test set and print the Mean Squared Error (MSE) and R ² score (coefficient of determination) for each polynomial degree. f)Plot actual vs. predicted power values for all polynomial degrees on the same graph for comparison and label the plot accordingly. g)Create a plot showing MSE vs. polynomial degree. Which degree has the lowest MSE? Write in the comment. h)Print the coefficients and intercept for each trained polynomial regression model.		9	LIU 2025				
231	5	Use each model to predict power output for wind speeds of 5, 10, 15, and 20 m/s, and print the predictions. What does KNN stand for?	C	1		K-Neural Networks	K-Means Neighbours	k Nearest Neighbours	K-Cluster Neighbours
232	5	In the context of kNN, what does 'distance' refer to?	B	1		Geographical distance	Difference in attribute values	Time difference	None of these
233	5	What is the main advantage of kNN?	A	1		No assumptions about data	Makes assumptions about data	Prone to overfitting	None of these
234	5	What is the main disadvantage of kNN?	B	1	LIU 2025	No assumptions about data	Sensitive to irrelevant features and the scale of the data	Not prone to overfitting	None of these
235	5	What does kNN use to make decisions?	B	1		Splitting criteria like entropy or gini index	Distance measures like Euclidean or Manhattan	Similarity measures like cosine similarity	None of these
236	5	What does the following code snippet represent? from sklearn.neighbors import KNeighborsClassifier knn = KNeighborsClassifier(n_neighbors=5)	B	1		Training a decision tree classifier	Initializing a kNN classifier	Implementing logistic regression	Initializing a random forest classifier
237	5	What does the following code snippet accomplish? from sklearn.model_selection import train_test_split from sklearn.neighbors import KNeighborsClassifier X_train, X_test, y_train, y_test = train_test_split(features, labels, test_size=0.3, random_state=42) knn = KNeighborsClassifier(n_neighbors=5) knn.fit(X_train, y_train) predicted_labels = knn.predict(X_test)	B	1		Initializes a kNN classifier with 5 neighbors and trains it on the entire dataset	Splits the dataset into training and testing sets, initializes a kNN classifier with 5 neighbors, and trains it on the training set	Trains a decision tree classifier with Gini index as the criterion	Initializes a logistic regression classifier and evaluates its performance on the test set
238	5	What class from scikit-learn is used to create a KNN classifier?	B	1	LIU 2025	KNeighborsRegressor()	KNeighborsClassifier()	knn_classify()	nearest_neighbors()
239	5	What metric is used by default in KNeighborsClassifier() to calculate distance between data points?	C	1	LIU 2025	Manhattan distance	Chebyshev distance	Euclidean distance	Jaccard similarity
240	5	What is the primary task of the k-Nearest Neighbors algorithm?	A	1		Classification	Regression	Clustering	Dimensionality reduction
241	5	In kNN, the value of k represents:	C	1		The number of features in the dataset	The number of clusters in the dataset	The number of nearest neighbors to consider	The number of classes in the dataset
242	5	What does the "fit" method in scikit-learn's KNeighborsClassifier class do?	A	1		Trains the model	Evaluates the model	Preprocesses the data	Visualizes the data
243	5	Which parameter of the KNeighborsClassifier determines the number of neighbors to consider?	D	1	LIU 2025	neighbors	k_value	k_neighbors	n_neighbors
244	5	Which of the following scenarios is an example where K-Nearest Neighbors (KNN) algorithm is not suitable?	A	1		Image classification with high-resolution images.	Fraud detection in credit card transactions.	Sentiment analysis of text data.	Speech recognition for voice commands.
245	5	What is the primary criterion for a decision tree using entropy?	A	1	LIU 2024	Information Gain	Gini Index	Chi-Square	Reduction in Variance
246	5	What is entropy in the context of a decision tree?	D	1		A measure of impurity or disorder	A measure of similarity	A measure of distance	A measure of impurity or disorder
247	5	What does a decision tree do?	C	1		It makes decisions	It predicts continuous outcomes	It classifies data into different classes	None of these
248	5	What is the disadvantage of a decision tree?	A	1		Prone to overfitting	Prone to underfitting	Not sensitive to outliers	None of these
249	5	What does a decision tree use to make decisions?	A	1		Splitting criteria like entropy or gini index	Distance measures like Euclidean or Manhattan	Similarity measures like cosine similarity	None of these
250	5	What is the primary task of the Decision Tree algorithm?	A	1		Classification	Regression	Clustering	Dimensionality reduction
251	5	Which of the following measures is used to quantify the randomness in a decision tree?	B	1	LIU 2025	Variance	Entropy	Standard Deviation	Mean Absolute Error
252	5	What is the purpose of the following code snippet? from sklearn.tree import DecisionTreeClassifier dt_classifier = DecisionTreeClassifier(criterion='entropy')	C	1		Initializing a decision tree classifier with Gini index	Initializing a kNN classifier with Euclidean distance	Initializing a decision tree classifier with entropy as the criterion	Initializing a logistic regression classifier
253	5	How is accuracy calculated in the context of a confusion matrix?	A	1		(True Positives + True Negatives) / Total Predictions	True Positives / (True Positives + False Positives)	(True Positives + True Negatives) / Total Actual Positives	True Negatives / (True Negatives + False Negatives)
254	5	Which metric from the confusion matrix reflects the proportion of correctly classified negative instances?	C	1		Accuracy	Sensitivity	Specificity	Error Rate
255	5	Which metric from the confusion matrix focuses on the ability of the model to correctly identify positive instances?	B	1		Accuracy	Sensitivity	Specificity	Error Rate
256	5	How is specificity calculated in the context of a confusion matrix?	A	1	LIU 2025	True Negatives / (True Negatives + False Positives)	True Positives / (True Positives + False Negatives)	(True Positives + True Negatives) / Total Actual Positives	(True Positives + True Negatives) / Total Predictions
257	5	How is the error rate calculated from a confusion matrix?	D	1	LIU 2025	True Negatives / (True Negatives + False Positives)	(True Positives + True Negatives) / Total Actual Positives	True Positives / (True Positives + False Negatives)	(False Positives + False Negatives) / Total Predictions
258	5	Which metric from the confusion matrix focuses on the ability of the model to correctly identify negative instances?	C	1		Accuracy	Sensitivity	Specificity	Error Rate
259	5	How is specificity calculated in the context of a confusion matrix?	A	1		True Negatives / (True Negatives + False Positives)	True Positives / (True Positives + False Negatives)	(True Positives + True Negatives) / Total Actual Positives	(True Positives + True Negatives) / Total Predictions

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
260	5	What is the purpose of the following code snippet? from sklearn.metrics import confusion_matrix conf_matrix = confusion_matrix(true_labels, predicted_labels)	A	1		Evaluating the confusion matrix	Training a decision tree classifier	Implementing kNN algorithm	Tuning hyperparameters for a random forest classifier
261	5	What does the following code snippet accomplish? print(conf_matrix[0,0] / (conf_matrix[0, 0] + conf_matrix[0, 1]))	B	1		Calculating the sensitivity of the classifier	Calculating the specificity of the classifier	Evaluating the F1 score	Printing the accuracy score of the model
262	5	What is the purpose of the following code snippet? from sklearn.tree import DecisionTreeClassifier from sklearn.metrics import confusion_matrix dt_classifier = DecisionTreeClassifier(criterion='entropy') dt_classifier.fit(X_train, y_train) predicted_labels = dt_classifier.predict(X_test) conf_matrix = confusion_matrix(y_test, predicted_labels)	A	1		Initializes a decision tree classifier with entropy as the criterion and evaluates its performance using a confusion matrix	Initializes a kNN classifier with 3 neighbors and evaluates its performance using a confusion matrix	Initializes a logistic regression classifier and evaluates its performance using a confusion matrix	Trains a random forest classifier and evaluates its performance using a confusion matrix
263	5	Write Python code to train a kNN classifier using the following steps: Split the dataset X into training and testing sets with a test size of 0.3 and a random state of 42. Initialize a kNN classifier with 5 neighbors. Train the classifier on the training set. Make predictions on the test set. Calculate and print the accuracy score of the classifier.		3					
264	5	Write Python code to train a decision tree classifier with entropy as the criterion using the following steps: Initialize a Decision Tree classifier with entropy as the criterion. Train the classifier on the training set. Make predictions on the test set. Calculate and print the confusion matrix for the classifier.		3					
265	5	Write Python code to evaluate the performance of a classification model using the following steps: Import the necessary functions from sklearn.metrics. Calculate and print the classification report for the true labels and predicted labels. Calculate and print the accuracy score of the classifier.		5					
266	5	Using the Iris dataset (https://raw.githubusercontent.com/pdsinroza/python2/main/Iris.csv?token=GHSAT0AAAAACQ7ZNWMQ3U6FOFL3702JIPAZQ5CWZA), write Python code to perform the following tasks: Split the dataset into features (X) and labels (y). Split the features and labels into training and testing sets with a test size of 0.2 and a random state of 42. Initialize a kNN classifier with 3 neighbors. Train the classifier on the training set. Make predictions on the test set. Calculate and print the accuracy score of the classifier.		5					
267	5	You are tasked with using the k-Nearest Neighbors (kNN) algorithm to classify whether patients have diabetes or not based on certain diagnostic measurements. You have been provided with diabetes.csv file. The datasets consist of several medical predictor (independent) variables and one target (dependent) variable, Outcome. Independent variables include the number of pregnancies the patient has had, their BMI, insulin level, age, and so on. Also perform Model Performance Analysis using confusion matrix.		5					
268	5	Given the Breast Cancer Wisconsin (Diagnostic) dataset, the objective is to build a kNN classification model that accurately predicts whether a tumor is benign or malignant based on the diagnostic features provided. The model should be trained on a portion of the dataset and evaluated on another portion to assess its performance. The ultimate goal is to create a reliable classifier that can assist healthcare professionals in diagnosing breast cancer accurately and early. Use cancer.csv file for dataset.		5					
269	5	Given the credit card transaction dataset, the objective is to build a kNN classification model that accurately predicts whether a transaction is fraudulent or non-fraudulent based on the transaction features provided. The model should be trained on historical transaction data and evaluated on another portion of the dataset to assess its performance. The ultimate goal is to create a reliable classifier that can automatically detect fraudulent transactions and prevent financial losses for credit card companies and cardholders. Use card_transdata.csv for dataset.		5					
270	5	The task involves building a k-Nearest Neighbors (kNN) regression model to predict the Air Quality Index (AQI) based on the latitude and longitude coordinates of various countries. The dataset used for this task contains information about the AQI levels and geographic locations (latitude and longitude) of different countries. The AQI serves as an indicator of air quality, with higher values indicating poorer air quality and vice versa. Use AQI and Lat Long of Countries.csv for dataset.		5					
271	5	The task involves building a Decision Tree classifier to predict whether to play tennis based on weather conditions. The dataset used for this task is the PlayTennis dataset, which contains information about various weather attributes such as outlook, temperature, humidity, and wind, along with the corresponding decision to play tennis or not. Use PlayTennis.csv for dataset.		5					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
272	5	Imagine that you are a medical researcher compiling data for a study. You have collected data about a set of patients, all of whom suffered from the same illness. During their course of treatment, each patient responded to one of 5 medications, Drug A, Drug B, Drug c, Drug x and y. Part of your job is to build a model to find out which drug might be appropriate for a future patient with the same illness. The feature sets of this dataset are Age, Sex, Blood Pressure, and Cholesterol of patients, and the target is the drug that each patient responded to. It is a sample of multiclass classifier, and you can use the training part of the dataset to build a decision tree, and then use it to predict the class of a unknown patient, or to prescribe it to a new patient. Use drug200.csv for dataset.		7					
273	5	You have been provided with a dataset named students_performance.csv, which contains various features related to students' performance in exams. The dataset includes features such as gender, race/ethnicity, parental level of education, lunch, test preparation course, and scores in math, reading, and writing. Your task is to perform classification to predict whether a student will pass or fail based on these features. Steps to complete: 1.Load the Dataset: oLoad the dataset students_performance.csv into a pandas DataFrame. oDisplay the first few rows to understand the structure of the dataset and check the statistics. 2.Preprocessing: oConvert the scores in math, reading, and writing to a binary pass/fail label. Consider a score of 50 or above as pass(1) and below 50 as fail(0). oDrop the scores in math, reading, and writing. oCreate a new Column named 'Overall_Pass' and create a binary label: pass(1) if Student passes all heads (math, reading, and writing) and fail(0) if student has failed in even one head. oEncode categorical variables (gender, race/ethnicity, parental level of education, lunch, test preparation course) using one-hot encoding. oHandle any missing values if present. oSplit the dataset into features (x) and target (y) where the target is the pass/fail label for the math pass. oSplit the data into training (80%) sets using random state 10.		7					
274	5	Write Python code to train a Decision Tree Classifier using the following steps: a)Assume a sample dataset is already loaded and corresponding features and labels are already defined in the environment. b)Split the dataset into training and testing sets using 75% of the data for training and set random state of 10. c)Initialize an entropy-based Decision Tree Classifier with maximum depth of 3. Use this model to make predictions on the test data. d)Print confusion matrix and accuracy score of the classifier.		3	LUJ 2025				
275	5	Which of the following best describes the RandomForestClassifier algorithm?	B	1		A single decision tree model	An ensemble learning method that combines multiple decision trees	A clustering algorithm	A neural network model
276	5	The primary goal of RandomForestClassifier is to:	B	1		Predict continuous values	Classify data into different categories	Reduce dataset size	Perform dimensionality reduction
277	5	RandomForestClassifier reduces overfitting by:	B	1		Using a single deep tree	Combining predictions of multiple trees	Removing training data	Increasing bias
278	5	In RandomForestClassifier, each decision tree is trained on:	B	1		The entire dataset without variation	A bootstrap sample of the dataset	Only the test dataset	Only selected class labels
279	5	The final prediction in RandomForestClassifier is determined using:	B	1		Average of tree outputs	Majority voting	Gradient descent	Distance calculation
280	5	Which parameter controls the number of trees in RandomForestClassifier?	B	1		max_depth	n_estimators	criterion	random_state
281	5	Which module in Scikit-learn contains RandomForestClassifier?	C	1		sklearn.tree	sklearn.metrics	sklearn.ensemble	sklearn.preprocessing
282	5	What does RandomForestClassifier(n_estimators=100) do?	B	1		Trains the model immediately	Initializes a Random Forest classifier with 100 trees	Splits the dataset	Evaluates accuracy
283	5	The fit() method in RandomForestClassifier is used to:	B	1		Predict class labels	Train the model on training data	Calculate confusion matrix	Normalize data
284	5	Error Rate when FP=10 and FN=5 (Total=100)?	A	1		15%	20%	10%	25%
285	5	If False Positives increase, which metric decreases?	A	1		Specificity	Sensitivity	Recall	True Positives
286	5	Metric measuring correctly predicted positive cases?	B	1		Specificity	Sensitivity	Error Rate	Accuracy
287	5	Metric measuring correctly predicted negative cases?	C	1		Accuracy	Sensitivity	Specificity	Recall
288	5	Increasing number of trees generally:	B	1		Increases instability	Improves prediction stability	Reduces training data	Eliminates bias completely
289	5	Most common evaluation tool for RandomForestClassifier performance?	A	1		Confusion Matrix	PCA	Gradient Descent	Correlation Matrix
290	5	What does the term SVM stand for in Machine Learning?	B	1		Statistical Vector Machine	Support Vector Machine	Supervised Variable Model	Structured Vector Method
291	5	Support Vector Machine is primarily used for which type of problems?	C	1		Sorting problems	Clustering only	Classification and regression problems	Data visualization
292	5	The main objective of a Support Vector Machine classifier is to:	B	1		Minimize dataset size	Find the optimal decision boundary separating different classes	Increase bias	Remove outliers
293	5	In SVM, the decision boundary that separates classes is known as:	C	1		Margin	Support Line	Hyperplane	Separator Curve
294	5	Support vectors in SVM are defined as:	B	1		All data points in the dataset	Data points closest to the decision boundary	Randomly selected training samples	Only misclassified samples
295	5	In Support Vector Machine, the parameter C is known as the Regularization Parameter and it controls:	B	1		Number of classes in dataset	Trade-off between maximizing margin and minimizing classification errors	Size of dataset	Number of support vectors
296	5	If the value of C (Regularization Parameter) is set very high, the SVM model is more likely to:	C	1		Underfit the data	Ignore misclassification errors	Overfit the training data by penalizing errors heavily	Increase the margin significantly

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Sr. No.	unit_number	Question_text	answer	marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
297	5	If the value of C (Regularization Parameter) is set very low, the SVM model is more likely to:	B	1		Overfit the data	Underfit the data by allowing more classification errors	Eliminate support vectors	Increase variance
298	5	Which Scikit-learn class is commonly used to implement Support Vector Machine classification?	B	1		SVR	SVC	DecisionTreeClassifier	LogisticRegression
299	5	The predict() method in SVC is used to:	C	1		Train the model	Tune hyperparameters	Predict class labels for new data	Normalize the dataset
300	5	Support Vector Machine is considered a supervised learning algorithm because:	B	1		It works without labeled data	It requires labeled training data	It performs clustering automatically	It does not require training data
301	5	One key advantage of Support Vector Machine is that it:	B	1		Works only with small datasets	Performs well in high-dimensional spaces	Does not require parameter tuning	Always guarantees 100% accuracy
302	5	In SVM, misclassification of training samples is controlled by which parameter?	B	1		Margin	C (Regularization Parameter)	Hyperplane	Support vector count
303	5	One major advantage of Support Vector Machine over many other classifiers is that it:	B	1		Always guarantees zero error	Works efficiently in high-dimensional feature spaces	Requires very large datasets to function	Does not require model training
304	5	Which of the following statements about Support Vector Machine is correct?	C	1		SVM cannot handle high-dimensional data	SVM works only for binary classification	SVM aims to maximize the margin between classes	SVM does not require labeled data
305	5	In Support Vector Machine, which data points directly influence the position of the decision boundary?	C	1		All training samples	Only correctly classified samples	Support vectors	Randomly selected samples
306	5	Consider the following code snippet: from sklearn.svm import SVC model = SVC(C=1.0) model.fit(X_train, y_train) What is the purpose of this code?	B	1		To evaluate model accuracy	To train an SVM classification model	To split the dataset	To perform clustering
307	5	Consider the following code snippet: from sklearn.ensemble import RandomForestClassifier rf = RandomForestClassifier(n_estimators=100) rf.fit(X_train, y_train) What is the role of n_estimators=100?	B	1		Number of features	Number of trees in the forest	Number of training samples	Number of classes
308	5	Consider the following code snippet: rf = RandomForestClassifier(random_state=42) rf.fit(X_train, y_train) What is the purpose of random_state=42?	B	1		To increase model complexity	To ensure reproducible results	To reduce number of trees	To normalize dataset
309	5	Consider the following code snippet: rf.fit(X_train, y_train) y_pred = rf.predict(X_test) What is the purpose of the predict() method?	B	1		To train the model again	To generate predicted class labels	To calculate training loss	To remove overfitting
310	5	Breast Cancer Detection using SVM Using the cancer.csv dataset, build a classification model to predict whether a tumor is benign or malignant. Perform the following: Load and preprocess dataset. Separate features and target. Split dataset into training and testing sets. Train SVM classifier. Predict test dataset. Evaluate model accuracy.		6					
311	5	Diabetes Prediction using Random Forest You are given a dataset diabetes.csv containing medical records of patients. The dataset includes predictor variables such as BMI, Age, Insulin, Pregnancies, etc., and a target variable named Outcome. Your task is to: Load the dataset using pandas. Separate independent and dependent variables. Split dataset into training and testing sets. Train a Random Forest Classifier. Predict patient outcomes. Evaluate performance using: Confusion Matrix Accuracy Score		6					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
312	5	Iris Dataset Classification using SVM Using the Iris dataset, perform classification using Support Vector Machine by completing the following steps: Load the dataset. Separate features (X) and target labels (y). Split dataset into: Training set (80%) Testing set (20%) random_state = 42 Train an SVM classifier with C = 1. Predict the test dataset labels. Print the model accuracy.		6					
313	5	Random Forest Classifier Implementation Write Python code to implement a Random Forest Classifier using the following instructions: Initialize the classifier with: n_estimators = 100 random_state = 42 Train the model using training data. Predict class labels for test data. Evaluate the model by calculating: Confusion Matrix Accuracy score		6					
314	5	SVM Model Training and Evaluation Write a Python program to implement a Support Vector Machine (SVM) classifier using the following steps: Split the dataset into training and testing sets using: Test size = 0.3 Random state = 42 Initialize the SVM classifier with: C = 1.0 (Regularization parameter) Train the model using the training dataset. Predict the class labels for the test dataset. Calculate and print: Accuracy score Classification report		6					
315	6	Which of the following is the correct syntax for training a Keras model?	B	1		model.train(X_train, y_train, epochs=10, batch_size=32)	model.fit(X_train, y_train, epochs=10, batch_size=32)	model.train_on_data(X_train, y_train, epochs=10, batch_size=32)	model.fit_data(X_train, y_train, epochs=10, batch_size=32)
316	6	Which of the following is a way to prevent overfitting in a Keras model?	D	1	LJU 2025	Adding more layers	Increasing the learning rate	Decreasing the batch size	Adding dropout layers
317	6	Which of the following Keras layers can be used for image classification tasks?	A	1		Conv2D	LSTM	Dense	Dropout
318	6	What is the primary purpose of a Convolutional Neural Network (CNN)?	B	1	LJU 2025	Object detection	Image classification	Text generation	Reinforcement learning
319	6	Which layer type is typically used to extract local features in a CNN?	A	1	LJU 2025	Convolutional layer	Pooling layer	Fully connected layer	Activation layer
320	6	Which activation function is commonly used in the convolutional layers of a CNN?	A	1		ReLU (Rectified Linear Unit)	Sigmoid	Tanh (Hyperbolic Tangent)	Softmax
321	6	What is the purpose of the stride parameter in a convolutional layer?	A	1	LJU 2025	To control the step size of the convolution operation	To determine the size of the receptive field	To adjust the learning rate during training	None of the above
322	6	Which layer type is used to reduce the spatial dimensions in a CNN?	B	1		Convolutional layer	Pooling layer	Fully connected layer	Activation layer
323	6	Which layer type is responsible for applying non-linear transformations to the feature maps in a CNN?	D	1		Convolutional layer	Pooling layer	Fully connected layer	Activation layer
324	6	If I put a dropout parameter of 0.2, how many nodes will I lose?	A	1		20% of them	2% of them	20% of the untrained ones	2% of the untrained ones
325	6	If 0.05 dropout is used on a 300-node layer, how many nodes will be inactive during training?	A	1	LJU 2025	15	285	25	5

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326	6	If my data is sized 150x150, and I use Pooling of size 2x2, what size will the resulting image be?	D	1		300x300	148x148	149x149	75x75
327	6	If my Image is sized 150x150, and I pass a 3x3 Convolution over it, what size is the resulting image?	A	1		148x148	150x150	153x153	450x450
328	6	When exploring the graphs, the loss levelled out at about .75 after 2 epochs, but the accuracy climbed close to 1.0 after 15 epochs. What's the significance of this?	B	1		There was no point training after 2 epochs, as we overfit to the validation data	There was no point training after 2 epochs, as we overfit to the training data	A bigger training set would give us better validation accuracy	A bigger validation set would give us better training accuracy
329	6	Which is the correct line of code for adding Dropout of 20% of neurons using TensorFlow	C	1		tf.keras.layers.Dropout(20)	tf.keras.layers.DropoutNeurons(20)	tf.keras.layers.Dropout(0.2)	tf.keras.layers.DropoutNeurons(0.2)
330	6	You are required to write a Python program to implement a Convolutional Neural Network (CNN) using TensorFlow and Keras. Assume that the data preprocessing is already completed, and you have training and validation datasets ready to use. Your task is to create the CNN model, apply the necessary layers with specified parameters, compile the model, and fit it to the data. Follow the steps and specifications provided below: •Example of the Training and Validation datasets: Assuming x_train, y_train, x_valid, y_valid are already defined x_train and x_valid are the image data y_train and y_valid are the encoded labels of output data (converted into categorical data) • Import the necessary libraries. • Create a CNN model. • Add the layers to the model as specified: •First Convolutional Layer: oFilters: 75, Kernel Size: (3, 3), Activation: ReLU oInput Shape: (28, 28, 1) (assuming the images are 28x28 pixels with 1 color channels) •Batch Normalization Layer •Max Pooling Layer: Pool Size: (2, 2) •Second Convolutional Layer: oFilters: 50 oKernel Size: (3, 3) oActivation: ReLU •Batch Normalization Layer •Max Pooling Layer: Pool Size: (2, 2) •Dropout Layer: Rate: 0.25 •Third Convolutional Layer: oFilters: 25 oKernel Size: (3, 3) oActivation: ReLU •Batch Normalization Layer •Max Pooling Layer: Pool Size: (2, 2) •Flatten Layer •Dense Layer: oUnits: 512 oActivation: ReLU •Dropout Layer: Rate: 0.2 •Output Dense Layer: oUnits: 24 (assuming 24 classes for classification) oActivation: Softmax • Compile the model with the following specifications: oLoss: Categorical Crossentropy oMetrics: Accuracy • Fit the model to the training data with the following specifications: o Epochs: 5, Verbose: 1, Validation Data: Use the provided validation dataset		4	LIU 2024				
331	6	Design and implement an image classification system using Convolutional Neural Networks (CNN) with Transfer Learning. A pre-trained deep learning model is used to learn important image features, which helps in improving accuracy and reducing training time. The system trains the model on a given dataset, evaluates its performance, and predicts the class of new images.		7					
332	7	Which of the following libraries is used to parse data received from Open Weather Map API?	D	1		api	request	requests	json
333	7	What method in BeautifulSoup is used to find the first occurrence of a particular HTML element?	B	1		find_parent()	find()	select()	get_text()
334	7	What method in BeautifulSoup is used to find the ALL occurrence of a particular HTML element?	B	1		find_parent()	find_all()	select()	get_text()
335	7	how does one get the first header 1 tag after creating a soup object?	A	1		soup.h1	soup.header1	soup.h1[0]	soup.h1[1]

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
336	7	Which of the following finds all link tags?	D	1		all_links = soup.find('a')	all_links = soup.findall('a')	all_links = soup.findall('link')	all_links = soup.find_all('a')
337	7	Which format is constructed by nesting python dictionaries and lists as needed.	A	1		JSON	HTTP	HTML	XML
338	7	which function formats the BeautifulSoup parsed data, so that there each tag is on its own separate line with indentation.	A	1		prettyify()	beutify()	dump()	dumpS()
339	7	Which of the function of json library is used to print a json file with required indent?	B	1	UJU 2025	dummy()	dumps()	dummys()	dump()
340	7	Which of the following libraries is used to get response using api key from Open Weather Map api?	B	1	UJU 2025	api	requests	json	socket
341	7	How can you extract the text content of an HTML element using BeautifulSoup?	A	1		get_text()	get_content	text_content	extract_text()
342	7	What will be output of following code? <pre>from bs4 import BeautifulSoup html_doc = """ <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> <h2>This is an example HTML page</h2> <p> This is Python Tutorial for Beautiful Soup</p> <p>Learn HTML</p> <p>Learn CSS</p> </body> </html> """ soup = BeautifulSoup(html_doc, 'html.parser') print(soup.find('p').find('a')['href'])</pre>	D	1		Learn HTML	https://www.google.com/html/HTML-tutorials.php	This is Python Tutorial for Beautiful Soup	Error
343	7	What will be output of following code? <pre>from bs4 import BeautifulSoup html_doc = """ <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> <h2>This is an example HTML page</h2> <p> This is Python Tutorial for Beautiful Soup</p> <p>Learn HTML</p> <p>Learn CSS</p> </body> </html> """ soup = BeautifulSoup(html_doc, 'html.parser') print(soup.find('body').find_all('p')[1].get_text())</pre>	C	1		Learn CSS	This is Python Tutorial for Beautiful Soup	Learn HTML	Error
344	7	When scraping a website, you come across the following error: "HTTP Error 403: Forbidden". Which of the following could be the cause of this error?	A	1		The website has implemented measures to block web scraping activities	The website's server is temporarily down	The website's HTML structure has changed, causing the scraping script to fail	None of these
345	7	How can you extract the value of the href attribute from a link (<a> tag) using BeautifulSoup?	B	1		link.Collect ['href']	link.get('href')	link.href	link['href']

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Sr. No.	unit_number	Question_text	answer	marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
346	7	Find the correct syntax code from following codes.	A	1		import bs4 import requests url = 'https://indianexpress.com/' source = requests.get(url).text soup = bs4.BeautifulSoup(source,'html.parser') print(soup.prettify())	import bs4 import requests url = 'https://indianexpress.com/' source = requests.get(url).text soup = bs4.BeautifulSoup(source,'html.parser') print(soup.prettify())	import requests url = 'https://indianexpress.com/' source = requests.get(url).text soup = bs4.BeautifulSoup(source,'html.parser') print(soup.prettify())	import bs4 import requests url = 'https://indianexpress.com/' source = requests.get(url).text soup = bs4.BeautifulSoup(soup,'html.parser') print(soup.prettify())
347	7	Which function is used to send a GET request in Python?	B	1		requests.post()	requests.get()	requests.send()	requests.fetch()
348	7	What does HTTP status code 404 represent?	A	1		Not Found	OK	Unauthorized	Server Error
349	7	Which status code indicates successful request?	B	1		500	200	403	401
350	7	What is the default response format of most REST APIs?	C	1		XML	CSV	JSON	HTML
351	7	Which method converts API response into Python dictionary?	D	1		response.text()	response.content()	response.data()	response.json()
352	7	Which function converts Python object into JSON string?	A	1		json.dumps()	json.loads()	json.parse()	json.read()
353	7	What is pagination used for in APIs?	C	1		Encrypt data	Authenticate user	Divide large data into pages	Delete records
354	7	Which parameter is commonly used for pagination?	D	1		token	header	page	limit
355	7	HTTP headers are mainly used to:	A	1		Send additional information	Store database	Delete records	Increase speed
356	7	What operation is generally performed by the following code? requests.put(url, json={"name": "B"})	B	1	No	Create Resource	Update Resource	Delete Resource	Read Resource
357	7	Which function is used to send data to an API?	C	1		requests.get()	requests.delete()	requests.post()	requests.read()
358	7	Which HTTP method is used to update existing resources?	D	1		GET	POST	DELETE	PUT
359	7	Which header is commonly used for API authentication?	B	1		Content-Type	Authorization	Host	Server
360	7	Token-based authentication requires sending token in:	C	1		Body	URL	Header	Cookie only
361	7	What is rate limiting?	D	1		Unlimited requests	Encrypting JSON	Removing pagination	Limiting number of API requests
362	7	Which status code indicates too many requests?	A	1		429	200	404	500
363	7	REST APIs mainly use which protocol?	B	1		FTP	HTTP	SMTP	TCP
364	7	What is the output type of the following code? response = requests.get(url) print(type(response.json()))	B	1		str	dict	list	Response
365	7	What is the type of variable y? import json x = {"a":10} y = json.dumps(x)	B	1		dict	str	list	int
366	7	In the following request, where is the token sent? headers = {"Authorization": "Bearer abc123"} requests.get(url, headers=headers)	C	1		URL	Body	HTTP Header	Query Parameter
367	7	What does 'page=2' represent in the request below? requests.get("https://api.example.com/data?page=2")	C	1		Header	Token	Query Parameter	Status Code
368	7	What will happen if the key does not exist? data = response.json() print(data["name"])	C	1		TypeError	ValueError	KeyError	IndexError
369	7	Using Open Weather Map API, generate current air pollution data for Ahmedabad and extract detail of aqi.		3					
370	7	Using the Open Weather Map API, generate a 3 Hourly 5 Days weather forecast for Ahmedabad with all details in JSON format. Note: Request for all the data via API in metric units.		4					
371	7	Using the Open Weather Map API, find the location of ahmedabad		3					
372	7	Using the Open Weather Map API, find the wind_speed of ahmedabad		4					
373	7	Using the Open Weather Map API, generate a 3 hourly 5 days weather forecast for Ahmedabad with details like minimum temperature, maximum temperature, wind speed, humidity, and weather description. Display this data in the form of a Pandas data frame with the column names being date_time, min_temp, max_temp, wind_speed, humidity, and weather_description.		5					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
374	7	Write a Python program using beautiful soup to scrape all the news headlines in the div class "top news" from https://indianexpress.com/		5					
375	7	Write a program for web scrapping using BeautifulSoup to scrape the following details from the given link and make a data frame using that scraped data from the page in a given link. Link: https://www.politifact.com/factchecks You will find 30 news articles with fact checks on this page. You need to scrape the following details from all the articles and store that in a data frame. Statement of News, Date of News, Source of News.		6					
376	7	Write a program of web scrapping using BeautifulSoup to scrape the given data from the following link. https://editorial.rottentomatoes.com/guide/popular-movies/ On the above link, you'll find 30 Popular movies. Scrape the Movie Title and Rating of that particular movie and make a Dataframe of the same.		5					
377	7	Write a Python program to find the title tags from a given html document. html_doc = """" <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> This is an example HTML page Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc at nisi velit, aliquet iaculis est. Curabitur porttitor nisi vel lacus euismod egestas. In hac habitasse platea dictumst. In sagittis magna eu odio interdum mollis. Phasellus sagittis pulvinar facilisis. Donec vel odio volutpat tortor volutpat commodo. Donec vehicula vulputate sem, vel iaculis urna molestie eget. Sed pellentesque adipiscing tortor, at condimentum elit elementum sed. Mauris dignissim elementum nunc, non elementum felis condimentum eu. In in turpis quis erat imperdiet vulputate. Pellentesque mauris turpis, dignissim sed iaculis eu, euismod eget ipsum. Vivamus mollis adipiscing viverra. Morbi at sem eget nisi euismod porta. Learn HTML from w3resource.com Learn CSS from w3resource.com </body> </html> """"		3					
378	7	Write a Python program to retrieve all the paragraph tags from a given HTML document. html_doc = """" <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> This is an example HTML page Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc at nisi velit, aliquet iaculis est. Curabitur porttitor nisi vel lacus euismod egestas. In hac habitasse platea dictumst. In sagittis magna eu odio interdum mollis. Phasellus sagittis pulvinar facilisis. Donec vel odio volutpat tortor volutpat commodo. Donec vehicula vulputate sem, vel iaculis urna molestie eget. Sed pellentesque adipiscing tortor, at condimentum elit elementum sed. Mauris dignissim elementum nunc, non elementum felis condimentum eu. In in turpis quis erat imperdiet vulputate. Pellentesque mauris turpis, dignissim sed iaculis eu, euismod eget ipsum. Vivamus mollis adipiscing viverra. Morbi at sem eget nisi euismod porta. Learn HTML from w3resource.com Learn CSS from w3resource.com </body> </html> """"		3					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
379	7	Write a Python program to get the number of paragraph tags of a given html document. <code>html_doc = """ <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> <h2>This is an example HTML page</h2> <p> Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc at nisi velit, aliquet iaculis est. Curabitur porttitor nisi vel lacus euismod egestas. In hac habitasse platea dictumst. In sagittis magna eu odio interdum mollis. Phasellus sagittis pulvinar facilisis. Donec vel odio volutpat tortor volutpat commodo. Donec vehicula vulputate sem, vel iaculis urna molestie eget. Sed pellentesque adipiscing tortor, at condimentum elit elementum sed. Mauris dignissim elementum nunc, non elementum felis condimentum eu. In in turpis quis erat imperdiet vulputate. Pellentesque mauris turpis, dignissim sed iaculis eu, euismod eget ipsum. Vivamus mollis adipiscing viverra. Morbi at sem eget nisi euismod porta.</p> <p>Learn HTML from w3resource.com</p> <p>Learn CSS from w3resource.com</p> </body> </html> """</code>		3					
380	7	Write a Python program to extract the text in the first paragraph tag of a given HTML document. <code>html_doc = """ <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> <h2>This is an example HTML page</h2> <p> Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc at nisi velit, aliquet iaculis est. Curabitur porttitor nisi vel lacus euismod egestas. In hac habitasse platea dictumst. In sagittis magna eu odio interdum mollis. Phasellus sagittis pulvinar facilisis. Donec vel odio volutpat tortor volutpat commodo. Donec vehicula vulputate sem, vel iaculis urna molestie eget. Sed pellentesque adipiscing tortor, at condimentum elit elementum sed. Mauris dignissim elementum nunc, non elementum felis condimentum eu. In in turpis quis erat imperdiet vulputate. Pellentesque mauris turpis, dignissim sed iaculis eu, euismod eget ipsum. Vivamus mollis adipiscing viverra. Morbi at sem eget nisi euismod porta.</p> <p>Learn HTML from w3resource.com</p> <p>Learn CSS from w3resource.com</p> </body> </html> """</code>		3					
381	7	Write a Python program to find the length of the text of the first <h2> tag of a given html document <code>html_doc = """ <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> <h2>This is an example HTML page</h2> <p> Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc at nisi velit, aliquet iaculis est. Curabitur porttitor nisi vel lacus euismod egestas. In hac habitasse platea dictumst. In sagittis magna eu odio interdum mollis. Phasellus sagittis pulvinar facilisis. Donec vel odio volutpat tortor volutpat commodo. Donec vehicula vulputate sem, vel iaculis urna molestie eget. Sed pellentesque adipiscing tortor, at condimentum elit elementum sed. Mauris dignissim elementum nunc, non elementum felis condimentum eu. In in turpis quis erat imperdiet vulputate. Pellentesque mauris turpis, dignissim sed iaculis eu, euismod eget ipsum. Vivamus mollis adipiscing viverra. Morbi at sem eget nisi euismod porta.</p> <p>Learn HTML from w3resource.com</p> <p>Learn CSS from w3resource.com</p> </body> </html> """</code>		3					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
382	7	Write a Python program to find the text of the first <a> tag of a given html text. html_doc = """ <html> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"> <title>An example of HTML page</title> </head> <body> This is an example HTML page Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc at nisi velit, aliquet iaculis est. Curabitur porttitor nisi vel lacus euismod egestas. In hac habitasse platea dictumst. In sagittis magna eu odio interdum mollis. Phasellus sagittis pulvinar facilisis. Donec vel odio volutpat tortor volutpat commodo. Donec vehicula vulputate sem, vel iaculis urna molestie eget. Sed pellentesque adipiscing tortor, at condimentum elit elementum sed. Mauris dignissim elementum nunc, non elementum felis condimentum eu. In in turpis quis erat imperdiet vulputate. Pellentesque mauris turpis, dignissim sed iaculis eu, euismod eget ipsum. Vivamus mollis adipiscing viverra. Morbi at sem eget nisl euismod porta. Learn HTML from w3resource.com Learn CSS from w3resource.com </body> </html> """		3					
383	7	Write a Python program to interact with the following REST API: https://reqres.in/api/users The program should: Send a POST request to create a new user. Send JSON data containing name and job. Include a custom Authorization header token. Convert the API response to a Python dictionary(JSON).		5					
384	7	Write a Python program to send a POST request to the following REST API endpoint: https://jsonplaceholder.typicode.com/posts The program should: Send JSON data containing title, body, and userId. Include an Authorization token in the request header. Send the request using the requests library. Print appropriate messages based on the response status code: 201 → Data created successfully 400 → Bad request 401 → Unauthorized request Display the JSON response returned by the API.		5					
385	7	Write a Python program to extract all the URLs from the webpage python.org that are nested within tags from.		5					
386	7	Write a Python program to find all the h2 tags and list the first four from the webpage python.org.		5					
387	7	Find all the link tags and list the first ten from the webpage python.org		4					
388	7	Write a Python program to a list of all the h1, h2, h3 tags from the webpage python.org.		4					
389	7	Write a Python program to extract all the text from a given web page python.org.		3					
390	7	How to get the Daily News using Python. url='https://www.bbc.com/news'		5					
391	7	Find the title of the webpage. url = 'https://en.wikipedia.org/wiki/Python_(programming_language)'		4					
392	7	Find all the links on the page and print their URLs. url = 'https://en.wikipedia.org/wiki/Python_(programming_language)'		4					
393	7	Find the first paragraph on the page and print its text. url = 'https://en.wikipedia.org/wiki/Python_(programming_language)'		4					
394	7	Find all the headings on the page and print their text. url = 'https://en.wikipedia.org/wiki/Python_(programming_language)'		5					
395	7	Find the table on the page and its rows, Extract the data from each row, and print it url = 'https://en.wikipedia.org/wiki/List_of_countries_and_dependencies_by_population'		5					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																																																																																																																																																																																								
396	7	Find all the citation needed tags on the page.url = 'https://en.wikipedia.org/wiki/Python_(programming_language)' using BeautifulSoup.		5																																																																																																																																																																																													
397	7	You have been provided with html file named 'Scrape_this.html'. Scrape the mentioned data from the given html page using BeautifulSoup. <table><tr><th>Company Name</th><th>Current Price</th><th>Change %</th><th>Low</th><th>High</th><th>Value (₹ Cr.)</th><th>Volume (in 000's)</th></tr><tr><td>Vodafone Idea</td><td>7.75</td><td>1.44%</td><td>7.33</td><td>7.88</td><td>35.47</td><td>45,763.00</td></tr><tr><td>Reliance Power</td><td>15.01</td><td>-4.52%</td><td>14.82</td><td>15.62</td><td>31.44</td><td>20,947.33</td></tr><tr><td>Yes Bank</td><td>16.21</td><td>-0.12%</td><td>16.15</td><td>16.35</td><td>16.25</td><td>10,026.87</td></tr><tr><td>NHPC</td><td>45.80</td><td>0.57%</td><td>45.62</td><td>46.82</td><td>23.52</td><td>5,136.20</td></tr><tr><td>IDFC First Bank</td><td>82.17</td><td>1.11%</td><td>81.51</td><td>84.35</td><td>38.62</td><td>4,699.80</td></tr><tr><td>PNB</td><td>52.88</td><td>1.73%</td><td>51.83</td><td>53.23</td><td>21.10</td><td>3,990.91</td></tr><tr><td>Zee Entertainment</td><td>180.35</td><td>-2.67%</td><td>176.00</td><td>188.35</td><td>71.06</td><td>3,940.34</td></tr><tr><td>Tata Steel</td><td>113.90</td><td>-0.09%</td><td>112.85</td><td>114.85</td><td>36.42</td><td>3,197.60</td></tr><tr><td>NBCC (India)</td><td>40.86</td><td>-0.78%</td><td>40.66</td><td>41.93</td><td>10.55</td><td>2,581.07</td></tr><tr><td>Motherson Sumi Sys</td><td>85.90</td><td>5.84%</td><td>82.41</td><td>86.15</td><td>19.55</td><td>2,275.35</td></tr><tr><td>HDFC Bank</td><td>1635.80</td><td>2.00%</td><td>1611.50</td><td>1637.00</td><td>355.57</td><td>2,173.68</td></tr><tr><td>MRPL</td><td>76.83</td><td>-2.38%</td><td>75.70</td><td>79.00</td><td>15.41</td><td>2,005.35</td></tr><tr><td>REC</td><td>163.80</td><td>7.02%</td><td>160.70</td><td>168.70</td><td>27.37</td><td>1,670.93</td></tr><tr><td>Edelweiss Financial</td><td>51.05</td><td>-0.62%</td><td>50.70</td><td>53.55</td><td>8.27</td><td>1,619.51</td></tr><tr><td>L&T Finance Holdings</td><td>123.55</td><td>4.61%</td><td>121.75</td><td>126.85</td><td>19.63</td><td>1,589.17</td></tr></table> Above is the example of how your html file will look like. You need to scrape the data from the above table given in html file. You need to scrape the data of all the columns which are highlighted with a black box in above image. •You need to scrape Company Name, hyperlink ('href' link) of company name, Current Price, Value and Volume for all the companies given in the html page. And make a Dataframe combining all the scraped data using pandas. Your output Dataframe should look like below: <table><tr><th></th><th>Company Name</th><th>Company Link</th><th>Current Price</th><th>Value</th><th>Volume</th></tr><tr><td>0</td><td>Vodafone Idea</td><td>https://www.ndtv.com/business/stock/vodafone-idea</td><td>7.75</td><td>35.47</td><td>45,763.00</td></tr><tr><td>1</td><td>Reliance Power</td><td>https://www.ndtv.com/business/stock/reliance-power</td><td>15.01</td><td>31.44</td><td>20,947.33</td></tr><tr><td>2</td><td>Yes Bank</td><td>https://www.ndtv.com/business/stock/yes-bank-ltd</td><td>16.21</td><td>16.25</td><td>10,026.87</td></tr><tr><td>3</td><td>NHPC</td><td>https://www.ndtv.com/business/stock/nhpc-ltd</td><td>45.80</td><td>23.52</td><td>5,136.20</td></tr><tr><td>4</td><td>IDFC First Bank</td><td>https://www.ndtv.com/business/stock/idfc-first-bank</td><td>82.17</td><td>38.62</td><td>4,699.80</td></tr><tr><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td></tr><tr><td>190</td><td>MRF</td><td>https://www.ndtv.com/business/stock/mrf-ltd</td><td>100429.90</td><td>4.73</td><td>0.47</td></tr><tr><td>191</td><td>P&G Hygiene</td><td>https://www.ndtv.com/business/stock/procter-gamble</td><td>13961.40</td><td>0.52</td><td>0.37</td></tr><tr><td>192</td><td>3M India</td><td>https://www.ndtv.com/business/stock/3m-india-ltd</td><td>27161.25</td><td>0.82</td><td>0.30</td></tr><tr><td>193</td><td>Abbott India</td><td>https://www.ndtv.com/business/stock/abbott-india</td><td>22685.05</td><td>0.64</td><td>0.28</td></tr><tr><td>194</td><td>Bajaj Hold & Invest</td><td>https://www.ndtv.com/business/stock/bajaj-hold-ltd</td><td>7005.10</td><td>0.16</td><td>0.23</td></tr></table> 195 rows x 5 columns Note: While making a Dataframe store the data of 'Current Price' and 'Value' Columns in a Float Datatype. After creating a Dataframe, •Check for any Null values, remove rows if any. •Remove Duplicate rows if there are any. •Find and remove outliers from the 'Current Price' column. Then make a Simper Linear Regression Model using sklearn library, Where, X=column named 'Value' and Y=column named 'Current Price'. •Use 20% as test size while you split the data. •Find co-efficient and intercept of the model. •Find Mean Squared Error of the model. -->If you are unable to complete the scraping, you can also attempt Linear Regression by using the data from sample image shown in output format as well. Marks will be given accordingly.	Company Name	Current Price	Change %	Low	High	Value (₹ Cr.)	Volume (in 000's)	Vodafone Idea	7.75	1.44%	7.33	7.88	35.47	45,763.00	Reliance Power	15.01	-4.52%	14.82	15.62	31.44	20,947.33	Yes Bank	16.21	-0.12%	16.15	16.35	16.25	10,026.87	NHPC	45.80	0.57%	45.62	46.82	23.52	5,136.20	IDFC First Bank	82.17	1.11%	81.51	84.35	38.62	4,699.80	PNB	52.88	1.73%	51.83	53.23	21.10	3,990.91	Zee Entertainment	180.35	-2.67%	176.00	188.35	71.06	3,940.34	Tata Steel	113.90	-0.09%	112.85	114.85	36.42	3,197.60	NBCC (India)	40.86	-0.78%	40.66	41.93	10.55	2,581.07	Motherson Sumi Sys	85.90	5.84%	82.41	86.15	19.55	2,275.35	HDFC Bank	1635.80	2.00%	1611.50	1637.00	355.57	2,173.68	MRPL	76.83	-2.38%	75.70	79.00	15.41	2,005.35	REC	163.80	7.02%	160.70	168.70	27.37	1,670.93	Edelweiss Financial	51.05	-0.62%	50.70	53.55	8.27	1,619.51	L&T Finance Holdings	123.55	4.61%	121.75	126.85	19.63	1,589.17		Company Name	Company Link	Current Price	Value	Volume	0	Vodafone Idea	https://www.ndtv.com/business/stock/vodafone-idea	7.75	35.47	45,763.00	1	Reliance Power	https://www.ndtv.com/business/stock/reliance-power	15.01	31.44	20,947.33	2	Yes Bank	https://www.ndtv.com/business/stock/yes-bank-ltd	16.21	16.25	10,026.87	3	NHPC	https://www.ndtv.com/business/stock/nhpc-ltd	45.80	23.52	5,136.20	4	IDFC First Bank	https://www.ndtv.com/business/stock/idfc-first-bank	82.17	38.62	4,699.80	...	...	...	...	...	...	190	MRF	https://www.ndtv.com/business/stock/mrf-ltd	100429.90	4.73	0.47	191	P&G Hygiene	https://www.ndtv.com/business/stock/procter-gamble	13961.40	0.52	0.37	192	3M India	https://www.ndtv.com/business/stock/3m-india-ltd	27161.25	0.82	0.30	193	Abbott India	https://www.ndtv.com/business/stock/abbott-india	22685.05	0.64	0.28	194	Bajaj Hold & Invest	https://www.ndtv.com/business/stock/bajaj-hold-ltd	7005.10	0.16	0.23		9					
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398	7	Part-1: (4 marks) Write a python code to scrape data from the file 'imdb.html', the file contains the list of upcoming movies to be released in India in year 2023 and 2024. Scrape the data to get the movie name and year. Create a DataFrame having columns containing movie name and year. Expected Outcome:		9																																																																																																																																																																																													

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																																				
		<table><thead><tr><th></th><th>Movie</th><th>Year</th></tr></thead><tbody><tr><td>0</td><td>1920: Horrors of the Heart</td><td>2023</td></tr><tr><td>1</td><td>Dhoomam</td><td>2023</td></tr><tr><td>2</td><td>Shivaji Surathkal 2</td><td>2023</td></tr><tr><td>3</td><td>Maidaan</td><td>2023</td></tr><tr><td>4</td><td>Thandatti</td><td>2023</td></tr><tr><td>...</td><td>...</td><td>...</td></tr><tr><td>149</td><td>Nadada</td><td>2024</td></tr><tr><td>150</td><td>Devara</td><td>2024</td></tr><tr><td>151</td><td>Bade Miyan Chote Miyan</td><td>2024</td></tr><tr><td>152</td><td>Sheran Di Kaum Punjabi</td><td>2024</td></tr><tr><td>153</td><td>Thalapathy 68</td><td>2024</td></tr></tbody></table> 154 rows x 2 columns Part-2 (5 marks) The dataset provided in 'kc_house_data.csv' contains house sale prices for King County, which includes Seattle. It includes homes sold between May 2014 and May 2015Perform the following tasks : 1) Load the csv to a dataframe named 'house_survey'. 2) Display the first 5 rows of the dataframe. 3) Drop the columns "id" and "Unnamed: 0" 4) Check all the null values present in all the columns of the dataframe. 5) Fill the missing values of the column 'bedrooms' with the mean of the column. 6) Fill the missing values of the column 'bathrooms' with the mean of the column. 7) Use the Pandas method corr() to find the feature other than price that is most correlated with price and mention your answer as a comment. 8) Fit a linear regression model to predict the 'price' using the list of features: ["floors", "waterfront", "lat", "bedrooms" , "sqft_basement" , "view" , "bathrooms", "sqft_living15" , "sqft_above", "grade", "sqft_living"] 9) Consider 30% testing samples and use random state 10. 10) Find the Mean Squared Error.		Movie	Year	0	1920: Horrors of the Heart	2023	1	Dhoomam	2023	2	Shivaji Surathkal 2	2023	3	Maidaan	2023	4	Thandatti	2023	...	...	...	149	Nadada	2024	150	Devara	2024	151	Bade Miyan Chote Miyan	2024	152	Sheran Di Kaum Punjabi	2024	153	Thalapathy 68	2024							
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399	7	Write a python program to extract title, story and links as per given output from file named 'html1.html'. Required Output: The Dormouse's story Once upon a time there were three little sisters; and their names were Elsie, Lacie and Tillie; and they lived at the bottom of a well. [Elsie, Lacie, Tillie] (Print output wherever required)		4																																									
400	7	For the given fakepython html file, write a python program using BeautifulSoup library and perform following tasks - 1. Extract all Python related job titles and print them. 2. Extract all job titles, locations and companies and print them. Example of job title, location and company from fakepython html is given below –		4																																									

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		Senior Python Developer Payne, Roberts and Davis Stewartbury, AA 2021-04-08 Energy engineer Vasquez-Davidson Christopherville, AA 2021-04-08 Job title Company location																																																						
401	7	Write a Program of Web scraping using BeautifulSoup to scrape the given data from the given HTML file. In the given HTML file, you'll find 50 Movies. Scrape the Movie Title, Year and Rating of that particular movie and make a DataFrame of the same. Sample Snippet of Output: <table><thead><tr><th></th><th>Title</th><th>Year</th><th>Rating</th></tr></thead><tbody><tr><td>0</td><td>Succession</td><td>(2018–2023)</td><td>8.9</td></tr><tr><td>1</td><td>Spider-Man: Into the Spider-Verse</td><td>(2018)</td><td>8.4</td></tr><tr><td>2</td><td>Manifest</td><td>(2018–2023)</td><td>7.1</td></tr><tr><td>3</td><td>Barry</td><td>(2018–2023)</td><td>8.4</td></tr><tr><td>4</td><td>Yellowstone</td><td>(2018–2023)</td><td>8.7</td></tr><tr><td>5</td><td>The Rookie</td><td>(2018–)</td><td>8.0</td></tr><tr><td>6</td><td>Tom Clancy's Jack Ryan</td><td>(2018–2023)</td><td>8.0</td></tr><tr><td>7</td><td>Mayans M.C.</td><td>(2018–)</td><td>7.6</td></tr><tr><td>8</td><td>9-1-1</td><td>(2018–)</td><td>7.9</td></tr><tr><td>9</td><td>You</td><td>(2018–2024)</td><td>7.7</td></tr><tr><td>10</td><td>New Amsterdam</td><td>(2018–2023)</td><td>8.0</td></tr></tbody></table>		Title	Year	Rating	0	Succession	(2018–2023)	8.9	1	Spider-Man: Into the Spider-Verse	(2018)	8.4	2	Manifest	(2018–2023)	7.1	3	Barry	(2018–2023)	8.4	4	Yellowstone	(2018–2023)	8.7	5	The Rookie	(2018–)	8.0	6	Tom Clancy's Jack Ryan	(2018–2023)	8.0	7	Mayans M.C.	(2018–)	7.6	8	9-1-1	(2018–)	7.9	9	You	(2018–2024)	7.7	10	New Amsterdam	(2018–2023)	8.0		4				
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402	7	Using Open Weather Map API, generate current weather data for Delhi and extract detail of pressure, humidity, sea_level, visibility, timezone, sunset, description, speed. latitude and longitude of delhi is given below: lat= 28.6517178 lon= 77.2219388 API KEY : 9903d45b0c6a6259e1bcd8bb4e3daaec API call: https://api.openweathermap.org/data/2.5/weather?lat={lat}&lon={lon}&appid={API key}		4																																																				
403	7	For the given web page “Certified used Mercedes-Benz for sale.html” scrape the data of car Name, Mileage, Dealer name, Review Count and Price. Create DataFrame for that. Expected Output of dataframe.head is : <pre>NameMileageDealer NameReview CountPrice 02020 Mercedes-Benz CLA 250 Base 4MATIC28,744 mi.Mercedes-Benz of Lynnwood130\$35,995 12019 Mercedes-Benz AMG GT 53 Base25,771 mi.International Autos117\$79,995 22021 Mercedes-Benz AMG GLE 53 Base22,374 mi.John Sisson Motors38\$80,923 32022 Mercedes-Benz AMG CLA 45 Base 4MATIC10,595 mi.Mercedes-Benz of Santa Rosa30\$56,633 42021 Mercedes-Benz AMG GLE 53 Base33,622 mi.Mercedes-Benz of Rochester152\$76,995</pre>		5																																																				

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404	7	Using Open Weather Map API, generate current air pollution data for Delhi and extract detail of "nh3" lat=23.05 lon=14.05 API call request: http://api.openweathermap.org/data/2.5/air_pollution?lat={lat}&lon={lon}&appid={API key} Example of the API response: { "coord":[50, 50], "list":[{ "dt":1605182400, "main":{ "aqi":500 }, "components":{ "co":201.94053649902344, "no":0.01877197064459324, "no2":0.7711350917816162, "o3":68.66455078125, "so2":0.6407499313354492, "pm2_5":0.5, "pm10":0.540438711643219, "nh3":0.12369127571582794 } }] }		3																																																																																																				
405	7	To Scrape a Table From the below file using Beautiful Soup and make a data frame and print it. "Today 52 Week Low BSE_NSE Stocks Companies List – Ticker.html" Write a program to create a Model using linear regression to predict the 'Day Low Rs.' using the "price Rs.". Find coefficient, intercept, and mean squared error. Data Frame Output: <pre>1 print(df)</pre> <table><thead><tr><th>S.No.</th><th>Company</th><th>price Rs.</th><th>Day Low Rs.</th></tr></thead><tbody><tr><td>0</td><td>1 Rachana Infrastru</td><td>125.70</td><td>113.80</td></tr><tr><td>1</td><td>2 Tirupati Forge</td><td>7.80</td><td>7.75</td></tr><tr><td>2</td><td>3 Sera Investments&Fin</td><td>13.01</td><td>13.01</td></tr><tr><td>3</td><td>4 Viaz Tyres</td><td>46.35</td><td>45.00</td></tr><tr><td>4</td><td>5 Sellwin Traders</td><td>13.00</td><td>12.01</td></tr><tr><td>5</td><td>6 Soni Medicare</td><td>19.22</td><td>17.40</td></tr><tr><td>6</td><td>7 Sicagen India</td><td>24.35</td><td>24.00</td></tr><tr><td>7</td><td>8 Aspira Pathlab&Diagn</td><td>24.85</td><td>24.80</td></tr><tr><td>8</td><td>9 Patspin India</td><td>14.10</td><td>14.00</td></tr><tr><td>9</td><td>10 AG Universal</td><td>42.10</td><td>42.00</td></tr><tr><td>10</td><td>11 Arihant Foundn. &Hsg</td><td>39.65</td><td>38.50</td></tr><tr><td>11</td><td>12 Global Offshore Serv</td><td>7.85</td><td>7.75</td></tr><tr><td>12</td><td>13 GTN Textiles</td><td>19.50</td><td>19.50</td></tr><tr><td>13</td><td>14 GTN Inds</td><td>11.95</td><td>11.50</td></tr><tr><td>14</td><td>15 Vivanza Biosciences</td><td>8.33</td><td>8.33</td></tr><tr><td>15</td><td>16 TECIL Chem & Hydro</td><td>21.60</td><td>20.70</td></tr><tr><td>16</td><td>17 Milgrey Fin.&Invest</td><td>16.29</td><td>14.81</td></tr><tr><td>17</td><td>18 Zodiac-JRD-MKJ</td><td>31.95</td><td>31.50</td></tr><tr><td>18</td><td>19 Kanungo Financiers</td><td>5.35</td><td>4.85</td></tr><tr><td>19</td><td>20 CIL Nova Petro</td><td>16.05</td><td>15.25</td></tr><tr><td>20</td><td>21 Integ.Pro</td><td>8.81</td><td>8.81</td></tr><tr><td>21</td><td>22 Elango Inds</td><td>5.55</td><td>5.55</td></tr><tr><td>22</td><td>23 Voltaire Leasing</td><td>11.99</td><td>11.54</td></tr></tbody></table> Note: if you are not able to collect the data from web scraping. Make a data frame using a dictionary with a "price Rs." And "Day Low Rs."	S.No.	Company	price Rs.	Day Low Rs.	0	1 Rachana Infrastru	125.70	113.80	1	2 Tirupati Forge	7.80	7.75	2	3 Sera Investments&Fin	13.01	13.01	3	4 Viaz Tyres	46.35	45.00	4	5 Sellwin Traders	13.00	12.01	5	6 Soni Medicare	19.22	17.40	6	7 Sicagen India	24.35	24.00	7	8 Aspira Pathlab&Diagn	24.85	24.80	8	9 Patspin India	14.10	14.00	9	10 AG Universal	42.10	42.00	10	11 Arihant Foundn. &Hsg	39.65	38.50	11	12 Global Offshore Serv	7.85	7.75	12	13 GTN Textiles	19.50	19.50	13	14 GTN Inds	11.95	11.50	14	15 Vivanza Biosciences	8.33	8.33	15	16 TECIL Chem & Hydro	21.60	20.70	16	17 Milgrey Fin.&Invest	16.29	14.81	17	18 Zodiac-JRD-MKJ	31.95	31.50	18	19 Kanungo Financiers	5.35	4.85	19	20 CIL Nova Petro	16.05	15.25	20	21 Integ.Pro	8.81	8.81	21	22 Elango Inds	5.55	5.55	22	23 Voltaire Leasing	11.99	11.54		9				
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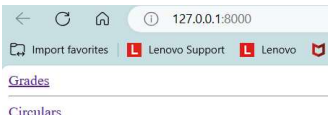
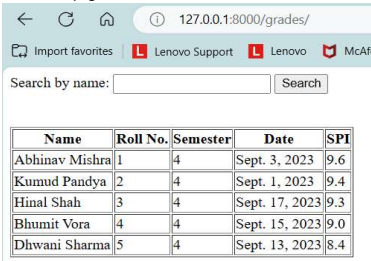
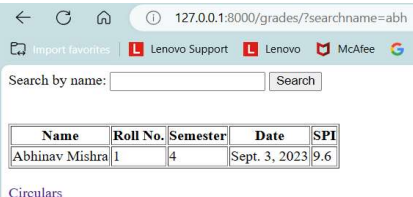
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406	7	Write a Program for Web scraping using BeautifulSoup to scrape the given data from the given HTML file named 'Popular_Quotes.html'. In the given HTML file, you'll find 30 Quotes. Scrape the Quote, Author and no of Likes of that particular Quote and make a DataFrame of the same. Sample Snippet of Output: <table><thead><tr><th></th><th>Quotes</th><th>Author</th><th>Likes</th></tr></thead><tbody><tr><td>0</td><td>"Be yourself, everyone else is already taken."</td><td>Oscar Wilde</td><td>172090</td></tr><tr><td>1</td><td>"I'm selfish, impatient and a little insecure....</td><td>Marilyn Monroe</td><td>162993</td></tr><tr><td>2</td><td>"So many books, so little time."</td><td>Frank Zappa</td><td>147069</td></tr><tr><td>3</td><td>"Two things are infinite: the universe and hum...</td><td>Albert Einstein</td><td>146082</td></tr><tr><td>4</td><td>"A room without books is like a body without a a...</td><td>Marcus Tullius Cicero</td><td>133363</td></tr><tr><td>5</td><td>"Be who you are and say what you feel, because...</td><td>Bernard M. Baruch</td><td>128599</td></tr><tr><td>6</td><td>"You've gotta dance like there's nobody watchi...</td><td>William W. Purkey</td><td>127116</td></tr><tr><td>7</td><td>"You know you're in love when you can't fall a...</td><td>Dr. Seuss</td><td>124987</td></tr><tr><td>8</td><td>"You only live once, but if you do it right, o...</td><td>Mae West</td><td>117240</td></tr><tr><td>9</td><td>"Be the change that you wish to see in the wor...</td><td>Mahatma Gandhi</td><td>113213</td></tr><tr><td>10</td><td>"In three words I can sum up everything I've l...</td><td>Robert Frost</td><td>108002</td></tr><tr><td>11</td><td>"If you want to know what a man's like, take a...</td><td>J.K. Rowling,</td><td>100983</td></tr><tr><td>12</td><td>"Don't walk in front of me... I may not followDo...</td><td>Albert Camus</td><td>92274</td></tr><tr><td>13</td><td>"If you tell the truth, you don't have to reme...</td><td>Mark Twain</td><td>90869</td></tr><tr><td>14</td><td>"Friendship ... is born at the moment when one...</td><td>C.S. Lewis,</td><td>87553</td></tr><tr><td>15</td><td>"I've learned that people will forget what you...</td><td>Maya Angelou</td><td>87398</td></tr></tbody></table>		Quotes	Author	Likes	0	"Be yourself, everyone else is already taken."	Oscar Wilde	172090	1	"I'm selfish, impatient and a little insecure....	Marilyn Monroe	162993	2	"So many books, so little time."	Frank Zappa	147069	3	"Two things are infinite: the universe and hum...	Albert Einstein	146082	4	"A room without books is like a body without a a...	Marcus Tullius Cicero	133363	5	"Be who you are and say what you feel, because...	Bernard M. Baruch	128599	6	"You've gotta dance like there's nobody watchi...	William W. Purkey	127116	7	"You know you're in love when you can't fall a...	Dr. Seuss	124987	8	"You only live once, but if you do it right, o...	Mae West	117240	9	"Be the change that you wish to see in the wor...	Mahatma Gandhi	113213	10	"In three words I can sum up everything I've l...	Robert Frost	108002	11	"If you want to know what a man's like, take a...	J.K. Rowling,	100983	12	"Don't walk in front of me... I may not followDo...	Albert Camus	92274	13	"If you tell the truth, you don't have to reme...	Mark Twain	90869	14	"Friendship ... is born at the moment when one...	C.S. Lewis,	87553	15	"I've learned that people will forget what you...	Maya Angelou	87398		9	LUJ 2024				
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407	7	From the given Popular_Quotes.html, load the file ensuring the utf-8 encoding and do the following task using BeautifulSoup library in python. 1.Extract the text from the <h1> tag and display it. 2.Retrieve the text of the <title> tag, then display it in reverse order. 3.What is the total number of <script> tags in the HTML? 4.Extract all tags listed under the first quote by Oscar Wilde and print the tag with the most characters. 5.Print the first 5 quotes along with their authors in the below given format: Example: Quote: "Be yourself; everyone else is already taken." Author: Oscar Wilde 6.Display the quotes that contain the word "accomplishment" in its text. 7.Extract the number of likes for each quote and display them in ascending order. 8.Find and print the names of authors who have more than one quote on the page. 9.Create a DataFrame containing the quote, author name, and number of likes.		9	LUJ 2025																																																																								
408	8	What is Django used for?	C	1		Machine learning	Game development	Web development	Data analysis																																																																				
409	8	What is Django in python?	A	1	LUJ 2023	A framework	A library	A function	A script																																																																				
410	8	What is the default database used in Django?	B	1		Oracle	SQLite	PostgreSQL	MySQL																																																																				
411	8	What is Django's template language used for?	C	1	LUJ 2025	Data validation	URL routing	Dynamic HTML generation	Object-Relational Mapping																																																																				
412	8	What is Django's admin app used for?	D	1	LUJ 2025	Generating dynamic HTML	Handling user authentication	Serving static files	Providing an interface for managing application data																																																																				
413	8	How do you run database migrations in Django?	A	1	LUJ 2023	python manage.py migrate	python migrate manage.py	django migrate	django manage.py migrate																																																																				
414	8	What is the purpose of the "urls.py" file in Django?	A	1	LUJ 2024	To store the project's URL configurations	To store project-level settings	To store app-level settings	To store the project's static files																																																																				
415	8	How do you make a Django model available for use in the admin interface?	A	1	LUJ 2024	By registering the model in the "admin.py" file of the app	By registering the model in the "settings.py" file of the project	By registering the model in the "models.py" file of the app	By registering the model in the "urls.py" file of the project																																																																				
416	8	What is the purpose of the "views.py" file in Django?	D	1		To store the project's URL configurations	To store the project's models	To store the project's static files	To store the project's views																																																																				
417	8	What is the purpose of the "forms.py" file in Django?	C	1		To store the project's models	To store the project's views	To store the project's forms	To store the project's URL configurations																																																																				

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Sr. No.	unit_number	Question_text	answer	Mark	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
418	8	What is the purpose of the "settings.py" file in Django?	B	1	LUJ 2024, LUJ 2025	To store app-level settings	To store project-level settings	To store the project's URL configurations	To store the project's static files
419	8	How do you run the development server in Django?	A	1		python manage.py runserver	python runserver manage.py	django runserver	django manage.py runserver
420	8	What is the purpose of the "migrations" folder in Django?	D	1		To store the project's views	To store the project's models	To store the project's static files	To store the project's database migrations
421	8	What is the purpose of the "__init__.py" file in Django?	C	1	LUJ 2023	To store the app's models	To store the app's views	To initialize the app	To store the app's forms
422	8	What are the features available in Django web framework?	D	1		Form handling	Templating	Admin Interface (CRUD)	All of the listed
423	8	What does {{ name }} mean in Django Templates?	C	1		{{ name }} will be the output.	It will be displayed as name in HTML.	The name will be replaced with values of Python variable.	None of the above
424	8	Which of the following is used if you need to deploy your project over WSGI?	A	1		wsgi.py	manage.py	models.py	settings.py
425	8	Which Django template tag is used to loop through a list of items?	A	1	LUJ 2023, LUJ 2024	{% for item in items %}	{% loop items %}	{% iterate items %}	{% list items %}
426	8	In which language is Django written?	A	1		PYTHON	PHP	JAVA	C++
427	8	Which Django command is used to create a new Django project?	C	1	LUJ 2025	StartProject	createproject	startproject	newproject
428	8	What is the primary purpose of the makemigrations command in Django?	A	1	LUJ 2023	To apply migrations to the database	To execute SQL queries for creating database tables	To create a superuser for the Django admin site	To run unit tests for your Django app
429	8	In Django, what is the primary purpose of the TEMPLATES_DIRS setting?	A	1	LUJ 2023, LUJ 2025	It defines the location(s) where Django looks for templates.	It lists all installed apps in the project.	It defines the URL patterns for a Django project.	It configures the project's settings.
430	8	Create Home page with logo and create about page ,put the link of about page in home page using Django.		6					
431	8	Make a student name search form by roll number using Django		10					
432	8	Create web page for book search by auther name or book name using django.		10					
433	8	Building an E-learning Platform using Django Create a platform for teachers to upload and schedule - notes, flowcharts, diagrams, videos, presentations, and others. Educators should be allowed to plan, organize, and display the curriculum for upcoming weeks for increased transparency.		9					
434	8	Creating a Contacts List Web App using Django •Create and deploy a new Django contacts list project •Create a new app in your contacts list project •Understand the model-view concept in Django and create a new view •Create a new model •Register your model in the admin app and access the model via admin •Create a view that displays all contacts data		9					
435	8	You are tasked with developing a simple Django web application to manage a library's book catalogue. The application should have two model classes, a view function, and necessary variables. Please note that the default code generated when creating a Django project/app is not to be included. Model Classes: 1.Book •Create a model class named Book with the following attributes: •title (CharField): A field for the title of the book, with a maximum length of 100 characters. •author (CharField): A field for the author's name, with a maximum length of 50 characters.		5					

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Sr. No.	unit_number	Question_text	answer	marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
		2. Borrower • Create another model class named Borrower with the following attributes: • name (CharField): A field for the name of the borrower, with a maximum length of 100 characters. • email (EmailField): A field for the borrower's email address, ensuring it's a valid email format. View Function: Write a view function named list_books that retrieves all books from the database and displays them in a template. Ensure that the template includes the book title, author, and publication year for each book. Styling for HTML code is not required. Necessary Variables: In your Django project settings (settings.py), assume you have configured the database settings correctly with an SQLite database and have created an app named library. Additionally, you can assume that Django's core settings, including the INSTALLED_APPS and URL routing, are properly set up already.							
436	8	Make a small website project using Django which has following functionality. 1). On the home page of the website, there should be 2 links called Grades and Circulars. 2). If you click on the Grades, you'll be redirected to a page where a table of marksheet will be displayed by extracting the data from Database stored in Django admin. Also, it should have functionality to search particular grades by passing name in search box. There will also be a link of circulars to redirect the page to circulars page. 3). If you click on a Circulars, the page should redirect to a page where circular will be displayed. There will also be a link of circulars to redirect the page to grades page. - You need to create a project named studentcorner. - You need to create an application called grades, which will contain page number 1 and 2 mentioned above. - You need to create another application called circulars, which will contain page number 3 mentioned above. - Create superuser and keep your roll no. as username and enrollment no. as password. (This is compulsory, otherwise marks will not be given.) Note: Data in the table of grades page needs to be displayed only by extracting it from the database stored in the admin page of your project. Static data in html file will not be considered. The home page should look like:  The Grades page should look like if all data are inserted in database:  Circulars If we search, 'abh' in search bar, it should display like below:  Circulars Circular page should look like below:		9					

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Fundamentals of Computer Science using Python - II Practice Book (SEM - IV)																																								
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		 127.0.0.1:8000/circulars/ Import favorites Lenovo Support Lenovo McA No Circulars to Display Grades Template for search box and table: <form action=""> <label>Search by name:</label> <input type="search" name="q" /> <button type="submit">Search</button> </form> <table border="1"> <tr> <th>Heading1 </th> <th>Heading2 </th> <th>Heading3 </th> <th>Heading4 </th> <th>Heading5 </th> </tr> <tr> <td>content1 </td> <td>content2 </td> <td>content3 </td> <td>content4 </td> <td>content5 </td> </tr> </table> If you want, you can add multiple rows same way. •No Marks will be given if your local server open with an error, no marks of logic will be given.																																						
437	8	Create a Python Django Project with your firstname, lastname, div and roll no. For example: Suppose your firstname is Chetan, lastname is Yadav, div is C8 and roll no. is 125 Hence the folder name formed should be : ChetanYadavC8125 The above example is an example you consider your example Follow the following steps: 1. Create an app called findmovie. This includes all steps like registering the new app and running the server. 2. Create a model named Movie with attributes title, year, description and director where title, description and director are character fields and year is an integer field. The maximum length of title, description and director are 100, 250 and 100 respectively. Migrate all sqlite tables. 3. Create a superuser with your RollNo like 125 and password should be lju123456. 4. Login to Django Admin Portal with this user and enter the following data in the Movie table: <table><tr><th>Movie Title</th><th>Director</th><th>Year</th><th>Description</th></tr><tr><td>Gadar 2</td><td>Anil Sharma</td><td>2023</td><td>Gadar 2 is a 2023 Indian Hindi-language period action drama film.</td></tr><tr><td>Bairangi Bhaijaan</td><td>Kabir Khan</td><td>2015</td><td>Bairangi Bhaijaan is a 2015 Indian Hindi-language comedy-drama film.</td></tr><tr><td>Pathaan</td><td>Siddharth Anand</td><td>2023</td><td>Pathaan is a 2023 Indian Hindi-language action thriller film.</td></tr><tr><td>Salaam Namaste</td><td>Siddharth Anand</td><td>2005</td><td>Salaam Namaste is a 2005 Indian romantic comedy film</td></tr><tr><td>83</td><td>Kabir Khan</td><td>2021</td><td>83 is a 2021 Indian Hindi-language biographical sports drama film</td></tr><tr><td>Genius</td><td>Anil Sharma</td><td>2018</td><td>Genius is a 2018 Indian Hindi-language romantic psychological action thriller film</td></tr><tr><td>Tanhaji: The Unsung Warrior</td><td>Om Raut</td><td>2020</td><td>Tanhaji: The Unsung Warrior is a 2020 Indian Hindi-language historical action film.</td></tr></table>	Movie Title	Director	Year	Description	Gadar 2	Anil Sharma	2023	Gadar 2 is a 2023 Indian Hindi-language period action drama film.	Bairangi Bhaijaan	Kabir Khan	2015	Bairangi Bhaijaan is a 2015 Indian Hindi-language comedy-drama film.	Pathaan	Siddharth Anand	2023	Pathaan is a 2023 Indian Hindi-language action thriller film.	Salaam Namaste	Siddharth Anand	2005	Salaam Namaste is a 2005 Indian romantic comedy film	83	Kabir Khan	2021	83 is a 2021 Indian Hindi-language biographical sports drama film	Genius	Anil Sharma	2018	Genius is a 2018 Indian Hindi-language romantic psychological action thriller film	Tanhaji: The Unsung Warrior	Om Raut	2020	Tanhaji: The Unsung Warrior is a 2020 Indian Hindi-language historical action film.		10				
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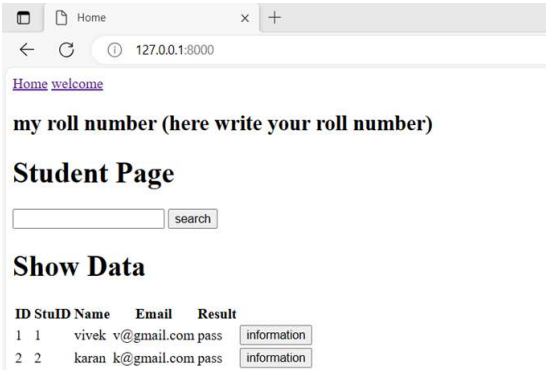
Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																							
		5. On the home page, display the following HTML form in a file named moviefind.html: <pre><h1> Home Page </h1> <form> <label>Title</label>
 <input type="text" name="title">
 <label>Year</label>
 <input type="text" name="year">
 <label>Director</label>
 <input type="text" name="director">
 <button type="submit"> Submit </button> </form></pre> This includes all like creating url for home page. 6. Make necessary adjustments to your code to let the user search for movie from this table by filling the HTML form. Filter your data on the basis for what the user filled in the HTML Form. Assume that the user will fill at least one field from the HTML form. And you can keep upto two fields in the form empty. Display the filtered movie data on your home page, below the form.																														
438	8	Create a Python Django Project with your firstname and lastname and follow the following steps: 1.Create an app called findmusic. 2.On the home page, display the following HTML form in a file name musicfind.html: <pre><form> <h1> Welcome </h1> <h2> Let's search your Music </h2> <input id="song" class="input" type="text" placeholder=" " /> <label for="song"> Song </label> <input id="artist" type="text" placeholder=" " /> <label for="artist"> Artist </label> <input id="year" type="text" placeholder=" " /> <label for="year"> Year </label> <input id="album" type="text" placeholder=" " /> <label for="album"> Album </label> <button type="submit"> Submit </button> </form></pre> 3.Create a model named Music with attributes song,artist,year and album where song,artist and album are text fields and year is a numeric field. 4.Migrate all the sqllite tables. 5.Create a superuser with your enrollment number last 4 digit as the username and password django123456 6.Log into the Django Admin Portal with this user and enter the following data in the Music Table <table><tr><th>Song</th><th>Artist</th><th>Year</th><th>Album</th></tr><tr><td>Calm Down</td><td>Selena Gomez</td><td>2022</td><td>Midnight Vibes</td></tr><tr><td>Ice Cream</td><td>Selena Gomez</td><td>2020</td><td>The album</td></tr><tr><td>People You Know</td><td>Selena Gomez</td><td>2020</td><td>Rare</td></tr><tr><td>Light Switch</td><td>Charlie Puth</td><td>2022</td><td>Charlie</td></tr><tr><td>We Don't talk Anymore</td><td>Charlie Puth</td><td>2016</td><td>Nine Track Mind</td></tr></table>	Song	Artist	Year	Album	Calm Down	Selena Gomez	2022	Midnight Vibes	Ice Cream	Selena Gomez	2020	The album	People You Know	Selena Gomez	2020	Rare	Light Switch	Charlie Puth	2022	Charlie	We Don't talk Anymore	Charlie Puth	2016	Nine Track Mind		9				
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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																																																																																							
439	8	Build a Django project using following guidelines. Write code for the following task: 1.Create a Django Project: Name of project is myproject 2.Create a Django App: Name of app is myapp 3. Define a Model: In myapp/models.py, define a simple model using name and description of project. 4. Create Migrations and Apply Them: 5. Create an Admin Interface: In myapp/admin.py, register your model to make it accessible in the Django admin interface: 6. Create Views and Templates: Create views in myapp/views.py and templates in myapp/templates/ to handle the display of your data. 7. Define URL Patterns: In myapp/urls.py, define URL patterns to map to your views. 8. Include App URLs in Project URLs: In myproject/urls.py, include the app's URL patterns. 9. Run the Development Server:		5																																																																																												
440	8	Creating a Django project to display data of cricket players as shown in table. Use the Django admin panel to add cricket player data. No need to display data of cricket players in table format only. You can display data of cricket players in any layout. Name of project should be CricketPlayers Name of app should be players Username and password should be LJENG <table><thead><tr><th>Player</th><th>Runs</th><th>Balls</th><th>4s</th><th>6s</th><th>SR</th><th>Team</th><th>Opposition</th></tr></thead><tbody><tr><td>RG Sharma</td><td>264</td><td>173</td><td>33</td><td>9</td><td>152.60</td><td>India</td><td>v Sri Lanka</td></tr><tr><td>MJ Guptill</td><td>237*</td><td>163</td><td>24</td><td>11</td><td>145.39</td><td>New Zealand</td><td>v West Indies</td></tr><tr><td>V Sehwag</td><td>219</td><td>149</td><td>25</td><td>7</td><td>146.97</td><td>India</td><td>v West Indies</td></tr><tr><td>CH Gayle</td><td>215</td><td>147</td><td>10</td><td>16</td><td>146.25</td><td>West Indies</td><td>v Zimbabwe</td></tr><tr><td>Fakhar Zaman</td><td>210*</td><td>156</td><td>24</td><td>5</td><td>134.61</td><td>Pakistan</td><td>v Zimbabwe</td></tr><tr><td>RG Sharma</td><td>209</td><td>158</td><td>12</td><td>16</td><td>132.27</td><td>India</td><td>v Australia</td></tr><tr><td>RG Sharma</td><td>208*</td><td>153</td><td>13</td><td>12</td><td>135.94</td><td>India</td><td>v Sri Lanka</td></tr><tr><td>SR Tendulkar</td><td>200*</td><td>147</td><td>25</td><td>3</td><td>136.05</td><td>India</td><td>v South Africa</td></tr><tr><td>CK Coventry</td><td>194*</td><td>156</td><td>16</td><td>7</td><td>124.35</td><td>Zimbabwe</td><td>v Bangladesh</td></tr><tr><td>Saeed Anwar</td><td>194</td><td>146</td><td>22</td><td>5</td><td>132.87</td><td>Pakistan</td><td>v India</td></tr></tbody></table>	Player	Runs	Balls	4s	6s	SR	Team	Opposition	RG Sharma	264	173	33	9	152.60	India	v Sri Lanka	MJ Guptill	237*	163	24	11	145.39	New Zealand	v West Indies	V Sehwag	219	149	25	7	146.97	India	v West Indies	CH Gayle	215	147	10	16	146.25	West Indies	v Zimbabwe	Fakhar Zaman	210*	156	24	5	134.61	Pakistan	v Zimbabwe	RG Sharma	209	158	12	16	132.27	India	v Australia	RG Sharma	208*	153	13	12	135.94	India	v Sri Lanka	SR Tendulkar	200*	147	25	3	136.05	India	v South Africa	CK Coventry	194*	156	16	7	124.35	Zimbabwe	v Bangladesh	Saeed Anwar	194	146	22	5	132.87	Pakistan	v India		10				
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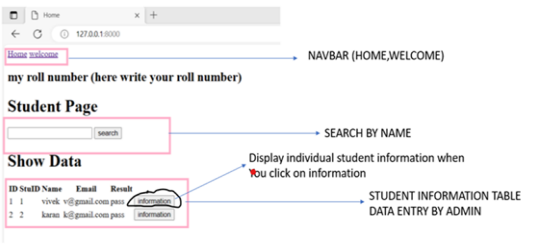
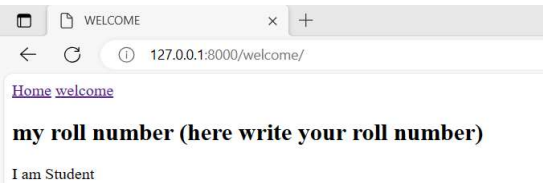
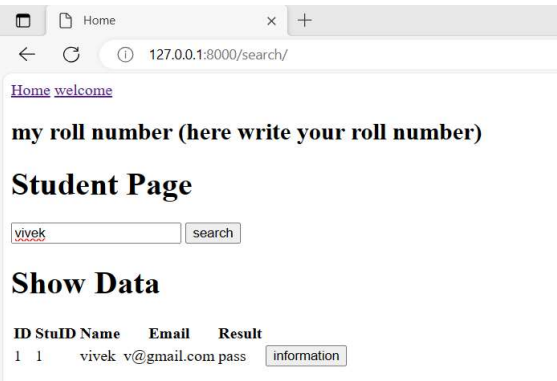
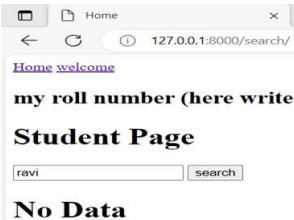
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Sr. No.	unit_number	Question_text	answ er	M ar ks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																		
441	8	•Create python Django project with name ‘musicproject’ •Create an app called musicapp •On homepage it will display songs list entered from admin portal. •Create home.html file in musicapp. •Code for home.html <pre><body> <h1>Music</h1> <h3>Enter Song </h3> <form action="" > <label for="song">Song:</label> <input type="text" name=" " >

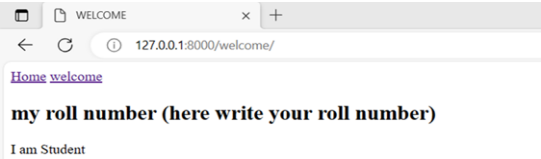
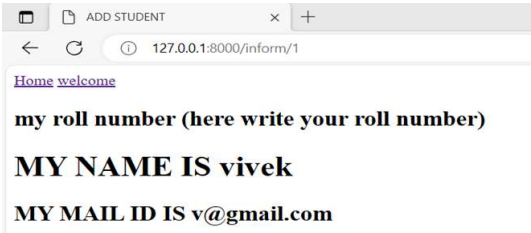
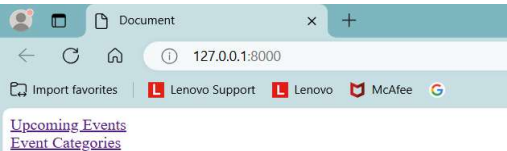
 <button type="submit" >Search</button> </form> </body></pre> •Create base.html file in musicproject. •Code for base.html <pre><!DOCTYPE html> <html> <head> <title>Music</title> </head> </html></pre> •Extends base.html file in to home html file. •Create model named Music with attributes song, artist, year. •Create super user with your enrollment number and password will be your roll number.(it is compulsory) •Log in to the django admin portal with this user and Enter the following data in Music table. necessary adjustment to your code to let user search for song from this database by song title on home page. <table><tr><th>song</th><th>artist</th><th>year</th></tr><tr><td>Ud jaa kale kawa</td><td>Alka Yagnik, Udit Narayan</td><td>2023</td></tr><tr><td>Dilwale Dulhania Le Jayenge</td><td>Lata Mangeshkar; S. P. Balasubraman</td><td>1995</td></tr><tr><td>All Izz Well</td><td>Sonu Nigam</td><td>2009</td></tr><tr><td>Radha kaise na jale</td><td>Asha bhosle</td><td>2001</td></tr><tr><td>Dil To Pagal Hai</td><td>Lata Mangeshkar</td><td>1997</td></tr></table>	song	artist	year	Ud jaa kale kawa	Alka Yagnik, Udit Narayan	2023	Dilwale Dulhania Le Jayenge	Lata Mangeshkar; S. P. Balasubraman	1995	All Izz Well	Sonu Nigam	2009	Radha kaise na jale	Asha bhosle	2001	Dil To Pagal Hai	Lata Mangeshkar	1997	•Make	9					
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Dil To Pagal Hai	Lata Mangeshkar	1997																									
442	8	Make a small website project using Django which has the following functionality.  EXPLAIN		10	LUJ 2023																						

EXPLAIN

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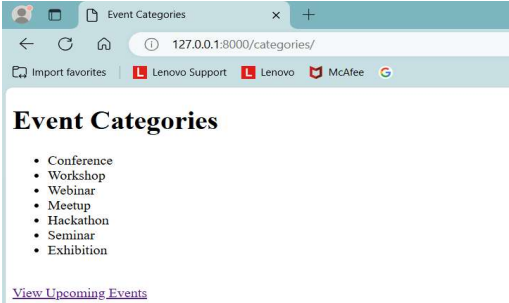
Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)	
		 1). On the home page of the website, there should be a navbar with a home and welcome link 2). If you click on the welcome, you'll be redirected to a page where you write I AM STUDENT  3). If you click on a Home, the page should redirect to a page where student data will be displayed. There will also be a (dynamic URL)link to information to redirect the page to the information page. -You need to create a project named project -You need to create an application called roll, -You need to create 4 function-based views: 1. student_search: It should have the functionality to search for a particular name by passing a name in the search box. If the student's name was not on the list then NO DATA 								
		2. Studentinfo: Display the student table on a home page. For data entry use Django administration.								

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
		 3. welcome: Display the I AM STUDENT  4. stuinformation: Display individual student information  -Create superuser and keep your roll no. as username and enrollment no. as password. (This is compulsory, otherwise marks will not be given.) Template for search box and table(only for your reference): <pre> <form action="" method="" "> <input type="text" name="" /> <input type="submit" name="" "> </form> <table > <tr> <th> Heading1 </th> <th> Heading2 </th> <th> Heading3 </th> <th> Heading4 </th> <th> Heading5 </th> </tr> <tr> <td> content1 </td> <td> content2 </td> <td> content3 </td> <td> content4 </td> <td> content5 </td> </tr> </pre>							
443		You are required to create a Django project named eventmanagemnt that includes two apps: events (for handling the events and their display) and categories (for listing the event categories). The project should meet the following specifications: 1.Home Page: oThe home page of the website should have 2 links named Upcoming Events and Event Categories. 							

L.J. Institute of Engineering & Technology, Ahmedabad.																																																																		
Fundamentals of Computer Science using Python - II Practice Book (SEM - IV)																																																																		
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		2.Upcoming Events Page (events app): oWhen you click on the Upcoming Events link, it should redirect you to a page where a table of upcoming events is displayed. oThe table should include the following columns: Event Name, Event Date, Location, and Category (e.g., Conference, Workshop, etc.). oThe data for this table should be extracted from the database stored in the Django admin interface with name Events and should be ordered by date in ascending order. (Enter the data in the database as shown in sample output) oThere should be a search bar that allows the user to search for events by event name. oThere will also be a link at the bottom of the page labelled Event Categories that will redirect the user to the Event Categories page of categories app. Upcoming Events Search by event name Search <table><tr><th>Event Name</th><th>Event Date</th><th>Location</th><th>Category</th></tr><tr><td>DjangoCon 2024</td><td>Sept. 15, 2024</td><td>Ahmedabad</td><td>Conference</td></tr><tr><td>Tech Meetup</td><td>Sept. 20, 2024</td><td>Rajkot</td><td>Meetup</td></tr><tr><td>Data Science Meetup</td><td>Sept. 21, 2024</td><td>Surat</td><td>Meetup</td></tr><tr><td>Startup Pitch Seminar</td><td>Sept. 23, 2024</td><td>Vadodara</td><td>Seminar</td></tr><tr><td>Innovation Expo</td><td>Sept. 26, 2024</td><td>Ahmedabad</td><td>Exhibition</td></tr><tr><td>Security Hackathon</td><td>Sept. 30, 2024</td><td>Surat</td><td>Hackathon</td></tr><tr><td>Machine Learning 101</td><td>Oct. 1, 2024</td><td>Vadodara</td><td>Webinar</td></tr><tr><td>Python Workshop</td><td>Oct. 5, 2024</td><td>Ahmedabad</td><td>Workshop</td></tr><tr><td>AI Hackathon</td><td>Oct. 10, 2024</td><td>Ahmedabad</td><td>Hackathon</td></tr><tr><td>Cloud Computing Webinar</td><td>Oct. 22, 2024</td><td>Ahmedabad</td><td>Webinar</td></tr></table> View Event Categories Upcoming Events Meet Search <table><tr><th>Event Name</th><th>Event Date</th><th>Location</th><th>Category</th></tr><tr><td>Tech Meetup</td><td>Sept. 20, 2024</td><td>Rajkot</td><td>Meetup</td></tr><tr><td>Data Science Meetup</td><td>Sept. 21, 2024</td><td>Surat</td><td>Meetup</td></tr></table> View Event Categories	Event Name	Event Date	Location	Category	DjangoCon 2024	Sept. 15, 2024	Ahmedabad	Conference	Tech Meetup	Sept. 20, 2024	Rajkot	Meetup	Data Science Meetup	Sept. 21, 2024	Surat	Meetup	Startup Pitch Seminar	Sept. 23, 2024	Vadodara	Seminar	Innovation Expo	Sept. 26, 2024	Ahmedabad	Exhibition	Security Hackathon	Sept. 30, 2024	Surat	Hackathon	Machine Learning 101	Oct. 1, 2024	Vadodara	Webinar	Python Workshop	Oct. 5, 2024	Ahmedabad	Workshop	AI Hackathon	Oct. 10, 2024	Ahmedabad	Hackathon	Cloud Computing Webinar	Oct. 22, 2024	Ahmedabad	Webinar	Event Name	Event Date	Location	Category	Tech Meetup	Sept. 20, 2024	Rajkot	Meetup	Data Science Meetup	Sept. 21, 2024	Surat	Meetup								
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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
	8	HTML Snippet For above webpage (Add appropriate DTL in it): <pre><html> <head> <title>Upcoming Events</title> </head> <body> <h1>Upcoming Events</h1> <form method="GET"> <input type="text" name="q" placeholder="Search by event name"> <button type="submit">Search</button> </form> <table border="1"> <thead> <tr> <th>Event Name</th> <th>Event Date</th> <th>Location</th> <th>Category</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>
 View Event Categories </body> </html></pre> 3.Event Categories Page (categories app): - oWhen you click on the Event Categories link, it should redirect you to a page where a list of event categories is displayed. - oThe categories should be extracted from the database stored in the Django admin interface with name Categories and displayed in order. - oThere will also be a link at the bottom of the page labelled Upcoming Events that will redirect the user to the Upcoming Events page of events app. 		10	LIU 2024				

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
		HTML Snippet For above webpage (Add appropriate DTL in it): <pre><html> <head> <title>Event Categories</title> </head> <body> <h1>Event Categories</h1>
 View Upcoming Events </body> </html></pre> 4.Admin Functionality: • Create a superuser with your roll number as the username and your enrollment number as the password. This superuser should be able to add and manage events and categories within the Django admin interface. • Ensure that all data for both pages is extracted dynamically from the database and not hardcoded in HTML.							
444		Project Statement: You are required to create a Django project named seminarsite that includes two apps: - seminars (for handling and displaying seminar details), and - departments (for listing the academic departments organizing the seminars). The project should meet the following specifications: 1. Home Page: • The homepage of the website should display two navigation links: - Upcoming Seminars - Departments 2. Upcoming Seminars Page (seminars app): • Clicking the Upcoming Seminars link should redirect to a page that displays a table of all upcoming seminars. • The table must contain the following columns: - Seminar Title - Seminar Date - Organizer - Department • The seminar data must be dynamically extracted from the database using the Seminars model and displayed in ascending order by seminar date. • A search bar should be provided to allow users to search seminars by their title. • At the bottom of the page, a link labelled 'Departments' should redirect the user to the Departments Page in the departments app.							

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Sr. No.	unit_ number	Question_text	answ er	M ar ks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
	8	HTML Snippet For above webpage (Add appropriate DTL in it): <pre><html> <head><title>Upcoming Seminars</title></head> <body> <h1>Upcoming Seminars</h1> <form method="GET"> <input type="text" name="q" placeholder="Search by seminar title"> <button type="submit">Search</button> </form> <table border="1"> <thead> <tr> <th>SeminarTitle</th><th>Date</th><th>Organizer</th><th>Department</th> </tr> </thead> <tbody> <tr> <td></td><td></td><td></td><td></td> </tr> </tbody> </table>
 View Departments </body> </html></pre> 3. Departments Page (departments app): - Clicking the Departments link should take the user to a page where all organizing departments are listed. - The department names should be retrieved dynamically from the database using the Departments model and displayed in a sorted order. - At the bottom of the page, a link labelled 'Upcoming Seminars' should redirect the user back to the Upcoming Seminars Page in the seminars app. HTML Snippet For above webpage (Add appropriate DTL in it): <pre><html> <head><title>Departments</title></head> <body> <h1>Departments</h1>
 View Upcoming Seminars </body> </html></pre> 4. Admin Functionality: - A Django superuser should be created using: - Username: Enrollment Number - Password: Roll Number - The superuser should be able to add, edit, and delete records for both Seminars and Departments through the Django admin interface. - All seminar and department data must be extracted dynamically from the database and not hardcoded in the HTML templates.	10	LUU 2025					
445	9	What is Django's forms module used for?	B	1	LUU 2025	URL routing	Data validation and form generation	Dynamic HTML generation	Object-Relational Mapping
446	9	What does CSRF stand for?	B	1	LUU 2024	Cross-Site Request Fraud	Cross-Site Request Forgery	Cross-Site Request Firewall	Cross-Site Response Forgery
447	9	How does Django protect against CSRF (Cross-Site Request Forgery) attacks?	C	1		By using a secret key	By using SSL/TLS	By using a unique token for each request	By using a fixed path
448	9	How does Django handle user authentication?	B	1		Through the use of a custom authentication backend	By providing a built-in authentication system	By relying on an external authentication service	By using the Django ORM
449	9	How to Ensure that you have installed Django successfully?	A	1		python -m django	python -m	python django	python -m Django
450	9	Which function is used to redirect users to a different URL in Django?	B	1		send_redirect	redirect	go_to	forward
451	9	What is the purpose of the authenticate function in Django?	B	1		It creates a new user.	It checks if a user is authenticated.	It logs a user in.	It logs a user out.
452	9	In Django Template Language, how can you check if a user is authenticated?	B	1	LUU 2023	user_logged_in	is_authenticated	check_authentication	is_logged_in

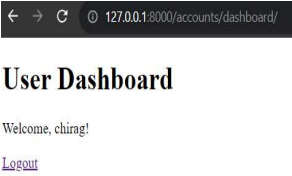
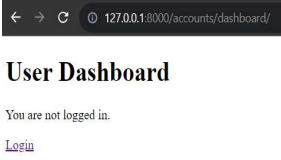
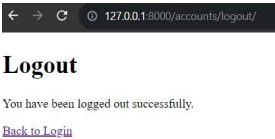
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Sr. No.	unit_number	Question_text	answer	marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
453	9	Which Django form is commonly used for user registration and account creation?	C	1	LUJ 2024	UserForm	RegisterForm	UserCreationForm	SignupForm
454	9	Which Django form is typically used for user login and authentication?	B	1		UserLoginForm	AuthenticationForm	UserLoginForm	AuthForm
455	9	How do you typically check for a unique username during user registration (sign-up) in Django?	C	1		Use the is_unique method on the username field.	Add a custom validation function to the username field.	Django automatically enforces unique usernames by default.	Use the unique attribute in the form field definition.
456	9	In Django, which function is responsible for verifying a user's credentials during the login process?	A	1	LUJ 2023, LUJ 2025	authenticate	create_user	user_login	validate_user
457	9	What does the 'logout' function do in Django's authentication system?	D	1		Logs a user in	Checks if a user is authenticated	Prevents Cross-Site Request Forgery (CSRF) attacks	Logs a user out
458	9	What is the primary function of the 'login' function in Django's authentication system during the login process?	C	1		It generates a new session ID.	It logs the user out.	It logs the user in.	It retrieves the user's profile information.
459	9	Which HTTP request method is commonly used for submitting form data in Django?	B	1	LUJ 2023, LUJ 2024	DELETE	POST	PUT	CONNECT
460	9	What is the primary purpose of Django's ORM (Object-Relational Mapping) in a web application built using the Django framework?	A	1	LUJ 2025	To define and interact with the database schema using Python code	To handle HTTP requests and responses	To handle asynchronous tasks in the background	To manage user authentication and authorization
461	9	Find the error in the following Django URL pattern configuration: from django.urls import path from . import views urlpatterns = [path('about/', views.about_view, name='about-page'), path('contact/', views.contact_view, name='contact-page'), path('products/<int:product_id>/', views.product_detail, name='product-detail'), path('categories/<str:category>/', views.category_view, name='category-page'), path('search/<str:keyword>/', views.search_view, name='search-page'), path('about/', views.contact_view, name='contact-page'), path('cart/<int:cart_id>/', views.cart_view, name='cart-page'), path('checkout/<int:order_id>/', views.checkout_view, name='checkout-page'),]	B	1		The path('about/', views.contact_view, name='contact-page') line is missing the leading forward slash (/) in the URL pattern	The path('about/', views.contact_view, name='contact-page') line is duplicated.	The path('search/<str:keyword>/', views.search_view, name='search-page') line should use <slug:keyword> instead of <str:keyword>	The path('cart/<int:cart_id>/', views.cart_view, name='cart-page') line should use <str:cart_id> instead of <int:cart_id>
462	9	By default, which HTTP method is protected by Django's CSRF protection?	B	1	LUJ 2025	PUT	POST	DELETE	GET
463	9	Which Django template tag is used for including the content of another template within a template file?	B	1		{% extend %}	{% include %}	{% block %}	{% including %}
464	9	How do you run database migrations in Django in a terminal?	A	1	LUJ 2025	python manage.py migrate	python migrate manage.py	django migrate	Django-admin migrate
465	9	Which file is kind of your project local django-admin for interacting with your project via command line?	A	1	LUJ 2024	manage.py	admin.py	urls.py	models.py
466	9	Build a Customer Relationship Management (CRM) App using Django need to cover following Contents in project: 1)Introduction 2)Installation and App Setup 3)Build Out the Basic App 4>Login Users 5)Logout Users 6)Register Users 7)View Records on Website 8)Individual Records 9)Add New Records		9					
467	9	Build an expense tracker app in Django •Set up your project •create database with user login, signup and logout functionalities •Add data •Design your report by fetching data		9					

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Sr. No.	unit_ number	Question_text	answ er	M ar ks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)																																									
468	9	Building a Blog Application using Django Key features of the project - 1. Creating and Retrieving blogs and authentication 2. Only admin can delete the posts 3. Change password and Contact Form		9																																														
469	9	Creating a Hotel Booking System using Django need to cover following Contents in project: Step 1: Install Django: Step 2: Create a New Django Project: Step 3: Create a New Django App: Step 4: Define Models: Step 5: Register Models: Step 6: Create Views and Templates: Step 7: Create URLs: Step 8: Create Forms: Step 9: Implement User Authentication: Step 10: Create User Registration View: Step 11: Create Navbar Template: Step 12: Integrate Rating System: Step 13: Create Reviews Page: Step 14: Include Navbar and Rating in Templates: Step 15: Handle Bookings and Payments: Step 16: Create superuser		9																																														
470	9	Building a User Login System for an Online Music Streaming Service In this project, you'll help Lushlyrics, a leading online music streaming service, enhance the security of its web application. You'll implement user authentication and authorization, working on a production-level website developed using the Django framework. Your primary goals are to secure the company's website and create a seamless customer registration and login experience. You'll develop an alternative version of the Lushlyrics site with signup/login functionality.		9																																														
471	9	Create views for login, signup and logout functionality in views.py file, assuming that you have html files named 'login.html', 'signup.html' and 'logout.html' respectively in templates folder in current app. Assume that form passes post request when login or signup button is pressed. -Following are the required modules, which needs to be imported for this functionality. from django.shortcuts import render from django.contrib.auth.models import User from django.contrib.auth.forms import AuthenticationForm, UserCreationForm from django.contrib.auth import login, logout, authenticate from django.shortcuts import redirect		5	LIU 2023, LIU 2024, LIU 2025																																													
472	9	You are required to develop a Django web application focused on cricket that includes the following features: 1. Cricket Database: •Create a Django project named 'cricstats' with a small database to manage information about cricket players. •The database should include the following fields for each player: •Player Name (CharField) •Country (CharField) •Batting Style (CharField) •Bowling Style (CharField) •Age (IntegerField) •Runs Scored (IntegerField) •Wickets Taken (IntegerField) <table><tr><th>Player Name</th><th>Country</th><th>Batting Style</th><th>Bowling Style</th><th>Age</th><th>Runs Scored</th><th>Wickets Taken</th></tr><tr><td>Virat Kohli</td><td>India</td><td>Right-handed</td><td>Right-arm medium</td><td>33</td><td>12169</td><td>4</td></tr><tr><td>Joe Root</td><td>England</td><td>Right-handed</td><td>Right-arm off</td><td>31</td><td>10491</td><td>32</td></tr><tr><td>Kane Williamson</td><td>New Zealand</td><td>Right-handed</td><td>Right-arm off</td><td>31</td><td>7115</td><td>30</td></tr><tr><td>Steve Smith</td><td>Australia</td><td>Right-handed</td><td>Leg break</td><td>32</td><td>7540</td><td>17</td></tr><tr><td>Babar Azam</td><td>Pakistan</td><td>Right-handed</td><td>Right-arm off</td><td>26</td><td>3859</td><td>1</td></tr></table>	Player Name	Country	Batting Style	Bowling Style	Age	Runs Scored	Wickets Taken	Virat Kohli	India	Right-handed	Right-arm medium	33	12169	4	Joe Root	England	Right-handed	Right-arm off	31	10491	32	Kane Williamson	New Zealand	Right-handed	Right-arm off	31	7115	30	Steve Smith	Australia	Right-handed	Leg break	32	7540	17	Babar Azam	Pakistan	Right-handed	Right-arm off	26	3859	1		10				
Player Name	Country	Batting Style	Bowling Style	Age	Runs Scored	Wickets Taken																																												
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L.J. Institute of Engineering & Technology, Ahmedabad. Fundamentals of Computer Science using Python - II Practice Book (SEM - IV)										
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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)	
		•Utilize the Django admin panel to set up and manage this database. Your Django admin should enable you to add, edit, and delete player entries. •The username should clearly be your roll no with password as ljet123, e.g. your Roll no is 200, the username should be 200 and password should be ljet123. 2. Web Pages: •Design a web application with multiple pages. Your application should include at least three primary pages: •Homepage: This page should display a list of cricket players from the database. Each player's entry should display their name. •Player Detail Page: When a user clicks on a player's name on the homepage, they should be directed to a detailed page that showcases all information about the selected player, including their name, country, batting style, bowling style, age, runs scored, and wickets taken. •Add Player Page: Create a form page that allows users to add new cricket players to the database. The form should include fields for all player attributes. •Add 2 players through this Add Player page. •Implement navigation links or a menu bar for easy access to these pages.								
473	9	For a Django project, create an application which enables the user to signup and login to access the dashboard, with ability to logout of the account. The detailed steps to be followed are as below – 1. Create a Django project called authedemoproject. 2. Create an app 'accounts' for the project. 3. Create views for handling signup, login, logout and dashboard for the accounts app importing following libraries (from django.contrib.auth.forms import UserCreationForm from django.contrib.auth import login, logout, authenticate from django.shortcuts import render, redirect) 4. Create necessary accounts app templates to render the views. Below are the required output snaps of different urls of the project - • The user must signup and login to redirect to the dashboard page. #accounts/signup.html ← → 🔄 📶 127.0.0.1:8000/accounts/signup/ Signup Username: Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only. Password: • Your password can't be too similar to your other personal information. • Your password must contain at least 8 characters. • Your password can't be a commonly used password. • Your password can't be entirely numeric. Password confirmation: Enter the same password as before, for verification. Signup ← → 🔄 📶 127.0.0.1:8000/accounts/login/ Login Username: Password: Login		10						

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
		• The dashboard page must have a link to logout page. • The dashboard must dynamically render the username with welcome message as shown below -  - Accessing the dashboard without logging in must render following template –  • The logout page must have a link to login page. <pre># accounts/logout.html 5. Setup applevel and project level urls to handle signup, login, logout and dashboard.</pre> 							
474	9	Create signupaccount function in views.py file in accounts app in django project named movieproject for signup functionality. There is signupaccount.html file in templates folder of accounts app for signup. After signup it should redirect to home page. views.py from django.shortcuts import render # Create your views here.		3					
475	9	Create urls.py file in accounts app in Django project named movieproject. Different Path available for url file are signup, login and logout with functions available in views.py as signupaccount, loginaccount and logoutaccount respectively. urls.py from django.urls import path		3					
476	10	What is the main purpose of a serializer in Django REST Framework?	B	1	No	Create database tables	Convert complex data to JSON	Render HTML	Manage URLs
477	10	Which class is used to automatically create serializer from model?	B	1	No	forms.ModelForm	serializers.ModelSerializer	models.Serializer	django.Form
478	10	What does serializer.is_valid() perform?	C	1	No	Saves data	Deletes data	Validates incoming data	Renders template
479	10	Which method is used to save validated serializer data?	C	1	No	serializer.commit()	serializer.push()	serializer.save()	serializer.store()
480	10	Where are model and fields defined in ModelSerializer?	C	1	No	ViewSet	class Meta	Meta class	Router
481	10	What is ViewSet in DRF?	B	1	No	HTML renderer	Class handling multiple CRUD operations	Database manager	URL mapper
482	10	Which class provides default CRUD operations?	C	1	No	APIView	GenericView	ModelViewSet	SimpleView
483	10	What is the purpose of DefaultRouter?	B	1	No	Create models	Automatically generate URL patterns	Validate serializer	Render HTML
484	10	Which method registers ViewSet with router?	C	1	No	router.add()	router.link()	router.register()	router.connect()
485	10	APIView is used when:	B	1	No	Auto CRUD required	Custom control over request handling	Only serializer needed	Only router required
486	10	What does JWT stand for?	B	1	No	Java Web Token	JSON Web Token	Joint Web Token	JavaScript Web Token
487	10	JWT consists of how many parts?	B	1	No	2	3	4	5
488	10	The three parts of JWT are:	B	1	No	Header Body Footer	Header Payload Signature	Token Key Hash	Header Key Signature

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Sr. No.	unit_number	Question_text	answer	marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
489	10	JWT token is generally sent in which header?	B	1	No	Content-Type	Authorization	Host	Server
490	10	Correct format of JWT in header is:	B	1	No	Token <token>	Bearer <token>	JWT <token>	Auth <token>
491	10	Role-based API restricts access based on:	B	1	No	IP address	User role	Server type	Database
492	10	Which permission allows only authenticated users?	B	1	No	AllowAny	IsAuthenticated	IsAdminUser	IsOwner
493	10	Which permission restricts access to admin users?	B	1	No	IsAuthenticated	IsAdminUser	AllowAny	IsStaff
494	10	Postman is used for:	B	1	No	Database design	API testing	Model training	Template rendering
495	10	Which HTTP method is used to update resource in Postman?	C	1	No	GET	POST	PUT	HEAD
496	10	Which status code indicates successful creation?	B	1	No	200	201	400	404
497	10	Which status code indicates unauthorized access?	C	1	No	200	403	401	500
498	10	API versioning is used to:	B	1	No	Increase speed	Maintain backward compatibility	Delete APIs	Secure APIs
499	10	Which is a best practice in API design?	B	1	No	Use random URLs	Use proper HTTP methods	Avoid status codes	Ignore authentication
500	10	Which attribute makes serializer field read-only?	B	1	No	write_only=True	read_only=True	required=False	editable=False
501	10	Which decorator defines custom action in ViewSet?	C	1	No	@api_view	@router	@action	@permission
502	10	Which method is overridden for custom validation?	B	1	No	check()	validate()	verify()	clean()
503	10	JWT token usually includes:	B	1	No	Database ID	Expiration time	HTML content	Session ID
504	10	Purpose of refresh token is:	B	1	No	Delete token	Generate new access token	Change password	Create user
505	10	Permission classes are defined in:	B	1	No	Serializer	ViewSet	Router	Model
506	10	JWT authentication class is added in:	B	1	No	models.py	settings.py	urls.py	forms.py
507	10	Which tab in Postman is used to add headers?	B	1	No	Body	Headers	Tests	Params
508	10	Delete request uses which method?	C	1	No	GET	POST	DELETE	PATCH
509	10	Router automatically generates which routes?	B	1	No	Template routes	CRUD routes	Static routes	Admin routes
510	10	Which status code indicates bad request?	C	1	No	200	201	400	404
511	10	Why should APIs use plural nouns?	B	1	No	Random naming	Represent collections	For styling	For database only
512	10	Purpose of JWT signature is:	B	1	No	Store data	Verify integrity	Render HTML	Route URL
513	10	Which serializer field is used for email validation?	C	1	No	CharField	IntegerField	EmailField	TextField
514	10	Which component enforces role-based access?	C	1	No	Router	Serializer	Permission classes	Template
515	10	Which class is used to return API response?	B	1	No	HttpResponse	Response	Render	Redirect
516	10	Validated data is accessed using:	C	1	No	serializer.data	serializer.json	serializer.validated_data	serializer.save
517	10	Why should APIs return proper status codes?	B	1	No	Random output	Clear client communication	Increase database speed	Hide data
518	10	Which component verifies JWT token?	C	1	No	Serializer	Router	Authentication class	Model
519	10	DRF mainly works on which architecture?	B	1	No	MVC	MVT	MVP	Layered
520	10	What will the following serializer do? class UserSerializer(serializers.ModelSerializer): class Meta: model = User fields = '__all__'	B	1	No	Create HTML form	Serialize all fields of User model	Delete user data	Create router
521	10	What happens if serializer.is_valid() returns False in the code? serializer = UserSerializer(data=request.data) serializer.is_valid()	C	1	No	Data saved automatically	Server crashes	Validation errors occur	Model deleted
522	10	What does the following class provide? class UserViewSet(viewsets.ModelViewSet): queryset = User.objects.all() serializer_class = UserSerializer	A	1	No	Automatic CRUD APIs	HTML pages only	Only database models	Only authentication
523	10	What will the following code do? router = DefaultRouter() router.register('users', UserViewSet)	B	1	No	Create serializer	Generate API URLs for UserViewSet	Delete routes	Create models
524	10	What is the purpose of this decorator? @api_view(['GET'])	D	1	No	Create router	Serialize data	Create database	Allow GET request for function-based API
525	10	What will this line return? return Response(serializer.data)	C	1	No	HTML page	Database object	JSON response	URL redirect
526	10	What does this header indicate? Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9	B	1	No	Database token	JWT authentication token	Router key	Serializer key
527	10	What does the following code check? if request.user.is_authenticated:	A	1	No	User authentication status	Database connection	Router configuration	Serializer validation
528	10	What does this permission class do? permission_classes = [IsAuthenticated]	D	1	No	Allow anyone	Allow only admin	Disable authentication	Allow only logged-in users
529	10	What does this return? return Response({'message': 'Success'})	A	1	No	JSON response message	Database record	Router URL	Serializer object

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Sr. No.	unit number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
530	10	Create a StudentSerializer using ModelSerializer with fields id, name, email, and marks. (Write Required codes of models.py & serializers.py) Requirements: Validate that marks must be between 0 and 100. Ensure email is unique. Return proper validation errors if conditions fail.		4					
531	10	Consider the following Django model: <pre>class Course(models.Model): title = models.CharField(max_length=100) duration = models.IntegerField()</pre> Create a Django REST Framework ViewSet that allows: Creating a course Retrieving courses Updating courses Deleting courses Use ModelViewSet. Files to Write - serializers.py - views.py		4					
532	10	Consider the model: <pre>class Product(models.Model): name = models.CharField(max_length=100) price = models.IntegerField()</pre> Write a function-based API view using <code>@api_view(['GET'])</code> that: Retrieves all products Returns data in JSON format Files to Write - views.py - serializers.py		5					
533	10	Consider the model: <pre>class Employee(models.Model): name = models.CharField(max_length=100) department = models.CharField(max_length=100)</pre> Create an API where: Admin users can create, update, and delete employees Normal users can only view employee records Use DRF permission classes. Files to Write - serializers.py - views.py		5					
534	10	Consider the model: <pre>class Movie(models.Model): title = models.CharField(max_length=100) rating = models.FloatField()</pre> Create a custom API endpoint inside a ViewSet that returns movies with rating greater than 8. Example endpoint: <code>/api/movies/top/</code> Use the <code>@action</code> decorator. Files to Write - views.py - serializers.py		5					

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Sr. No.	unit_number	Question_text	answer	Marks	Previous Year	option1 (A)	option2 (B)	option3 (C)	option4 (D)
535	10	Create the Online Bookstore Management System is a web-based application developed using the Django Framework and Django REST Framework that allows users to browse, search, and purchase books online. The system provides functionalities for administrators to add, update, and delete book records, while users can view book details and place orders. The application includes RESTful APIs for accessing book data, supports filtering and pagination of book listings, and implements secure authentication using JWT tokens. This project demonstrates the use of Django's MVT architecture, ORM, serializers, viewsets, routers, and API testing using Postman.		9					
536	10	Create the Student Performance Analysis Dashboard is a data-driven web application developed using Django, Django REST Framework, and machine learning techniques. The system collects student academic data and applies basic machine learning algorithms to analyze and predict student performance trends. The application allows administrators to manage student records through a web interface and RESTful APIs. It also provides graphical visualizations of performance metrics through an interactive GUI dashboard. The project demonstrates integration of data analysis, machine learning models, REST APIs, and secure authentication using JWT tokens within a Django-based system.		9					
537	10	Create the Online Event Management Platform is a web-based system developed using the Django framework and Django REST Framework that allows users to create, view, and manage events. The application enables event organizers to add event details, update schedules, and delete outdated events through a web interface and REST APIs. Users can search events based on date, location, and category. The system implements JWT-based authentication and role-based access control to ensure secure API access. This project demonstrates the implementation of Django models, serializers, viewsets, routers, API versioning, and REST API testing using Postman.		9					
538	10	Create the Movie Recommendation and Review System is a web-based application built using the Django Framework and Django REST Framework that enables users to explore movie listings, submit reviews, and receive personalized movie recommendations. The system allows administrators to manage movie data through CRUD operations while users can search and filter movies based on genres and ratings. The recommendation feature uses simple machine learning techniques to suggest movies based on user preferences. The application provides REST APIs for accessing movie data and uses JWT authentication for secure user access. This project demonstrates integration of web development, REST API design, and machine learning concepts.		9					